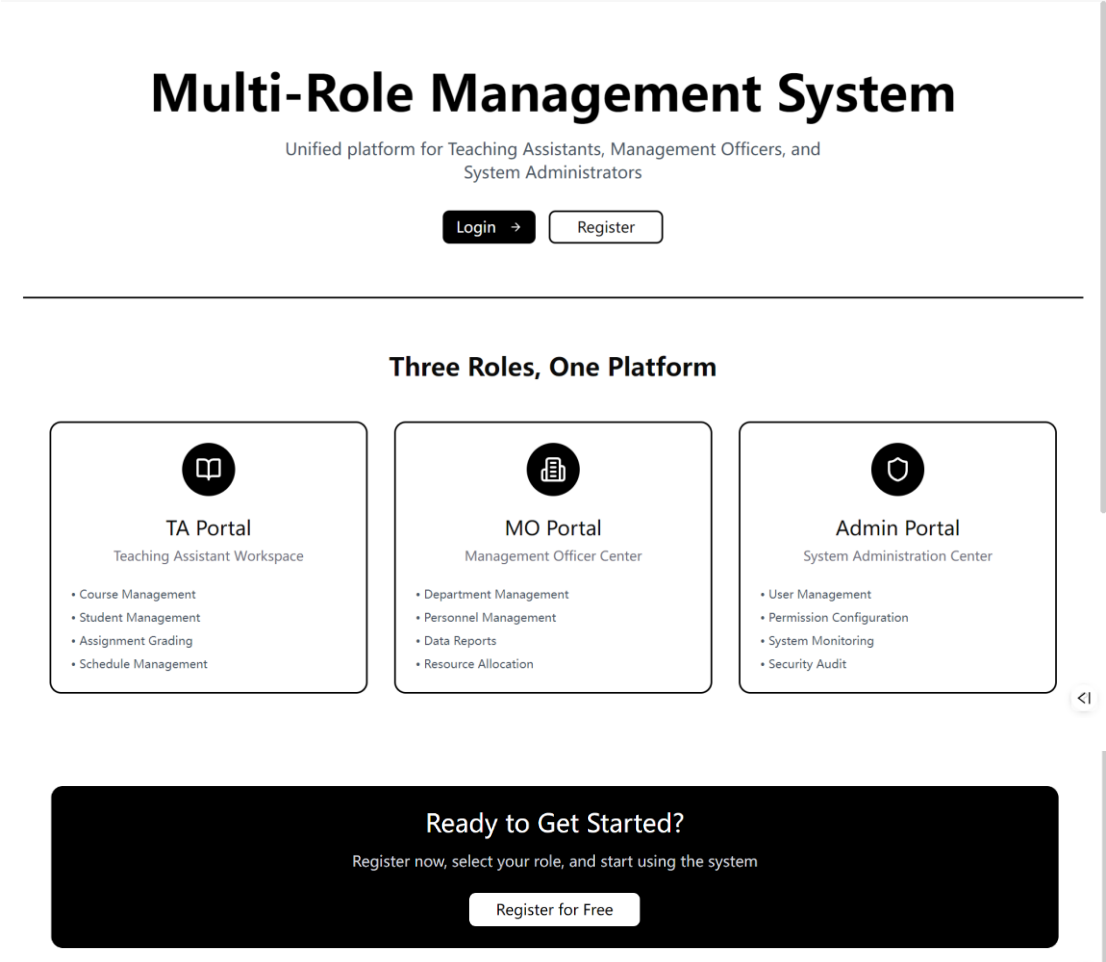
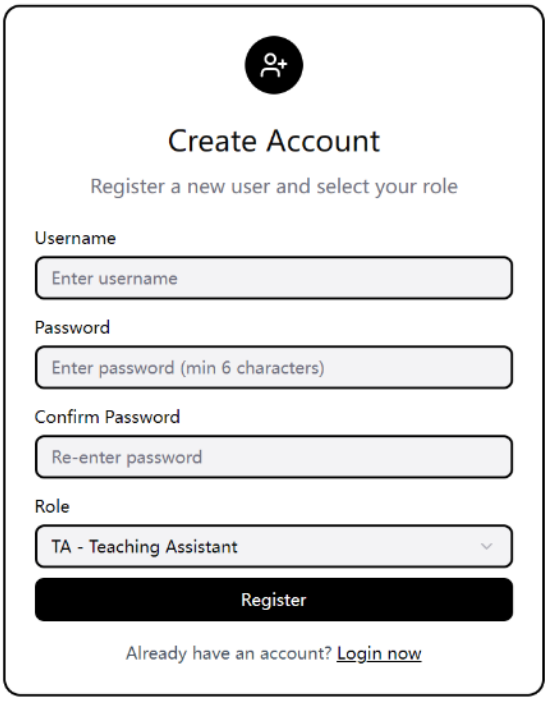
Authentication & Navigation module

(1) Homepage & Main Entry



The Authentication & Navigation module serves as the entry point of the entire system. Upon launching the application, the user is presented with the main interface. The homepage displays the "Multi-Role Management System" and offers two primary operational branches: Login and Register. Additionally, the page outlines the three core user roles supported by the platform: Teaching Assistants (TA), Management Officers (MO), and Administrators (Admin). The user must select an option here to proceed with the workflow.

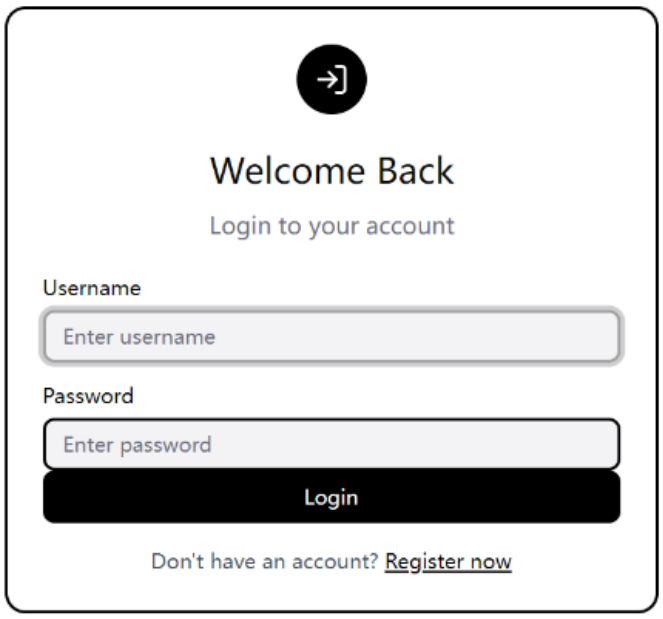
(2) Registration Workflow



When a new user selects "Register", the system directs them to the "Create Account" interface. In this form, the user is required to enter a username, set a password containing a minimum of 6 characters, and re-enter the password for confirmation. A crucial step in this process is role selection; the user must specify their identity (TA, MO, or Admin) via a dropdown menu. After clicking "Register", the system validates the uniqueness of the username and the validity of the password. If the validation is successful, the account is officially created ; if it fails, the system displays an error message.

(3) **Login Workflow**

For existing users who select "Login", the system displays the "Welcome Back" interface. The user inputs their registered username and password, then clicks the "Login" button to submit. The system then verifies these credentials. If the provided information is incorrect, an error message is shown on the screen.



(4) Role-Based Navigation

If the login authentication is successful, the system triggers a role-based navigation logic, redirecting the user to their designated workspace.

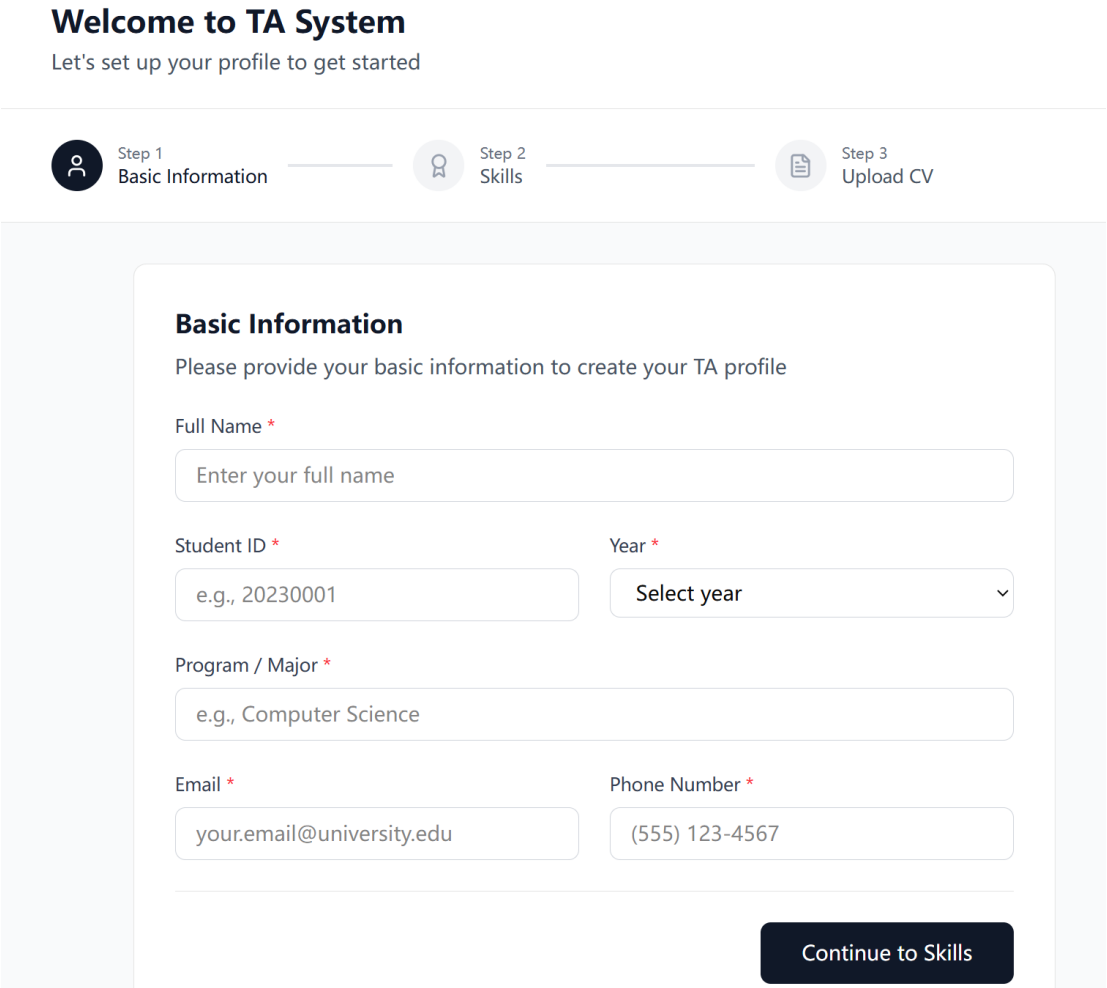
- TAs are routed to the TA Portal to handle tasks such as course management, grading, and scheduling.
- MOs are routed to the MO Portal, focusing on department and personnel management.
- Admins are routed to the Admin Portal, taking charge of user management and overall system monitoring.

(5) **Logout**

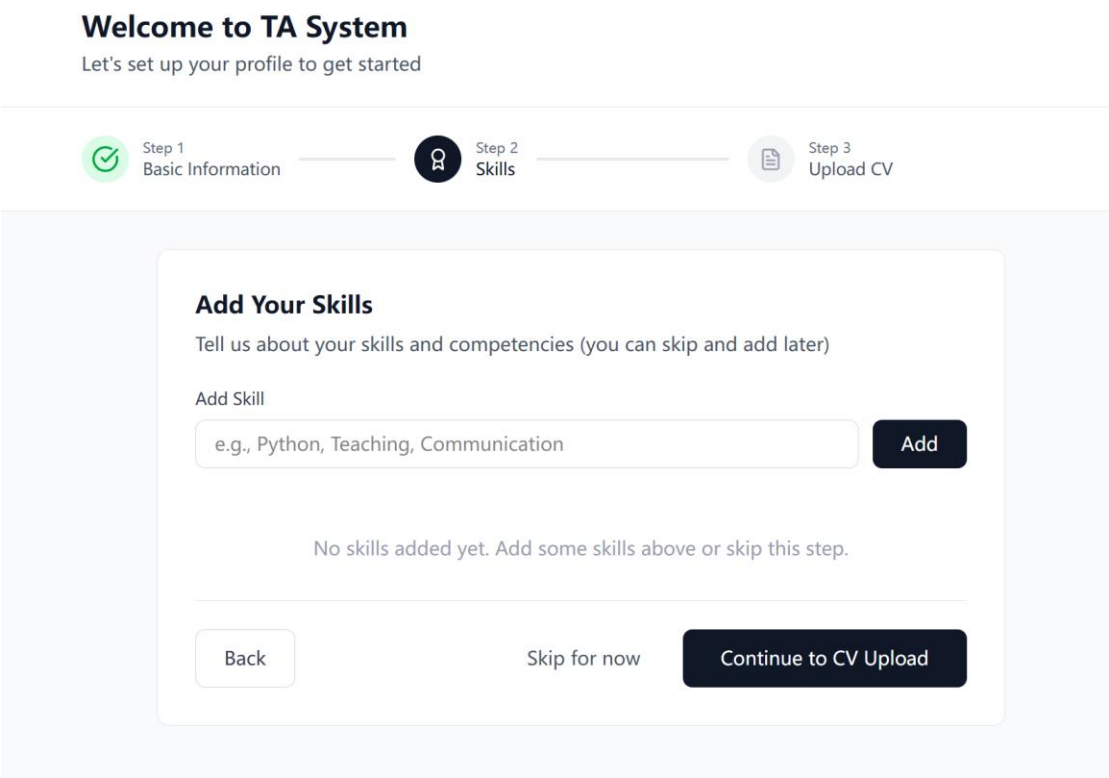
Once tasks are completed, users can log out of their accounts, at which point the system clears the current session to ensure security.

TA Profile Module

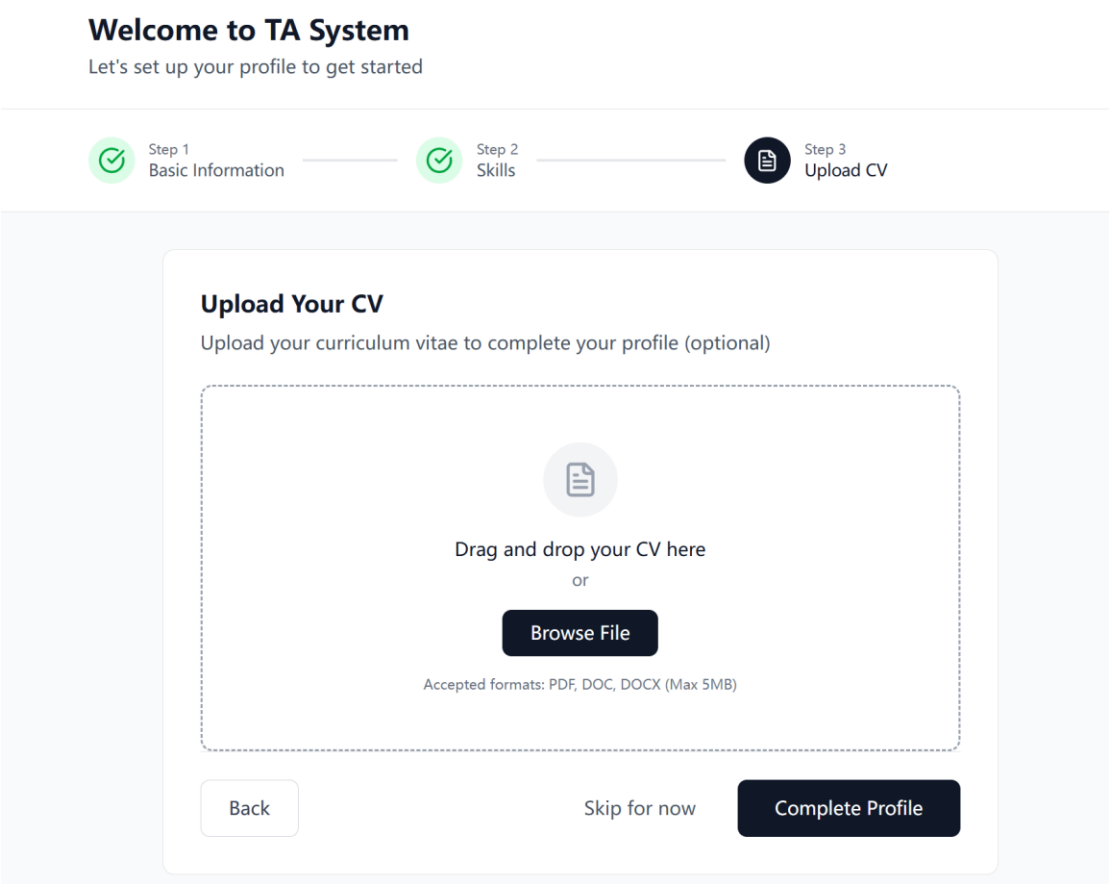
(1) Initial Profile Setup



When the user logs into the system for the first time, the system guides them into the initial setup process for their personal profile. This page serves as the starting point for the entire profile creation workflow, displaying the core basic information required to create a TA profile. In the center of the page, the user needs to enter their full name, student ID, program/major, year, email, and phone number. After confirming the information is correct, the user can click the "Continue to Skills" button at the bottom right of the page to proceed to the next step.

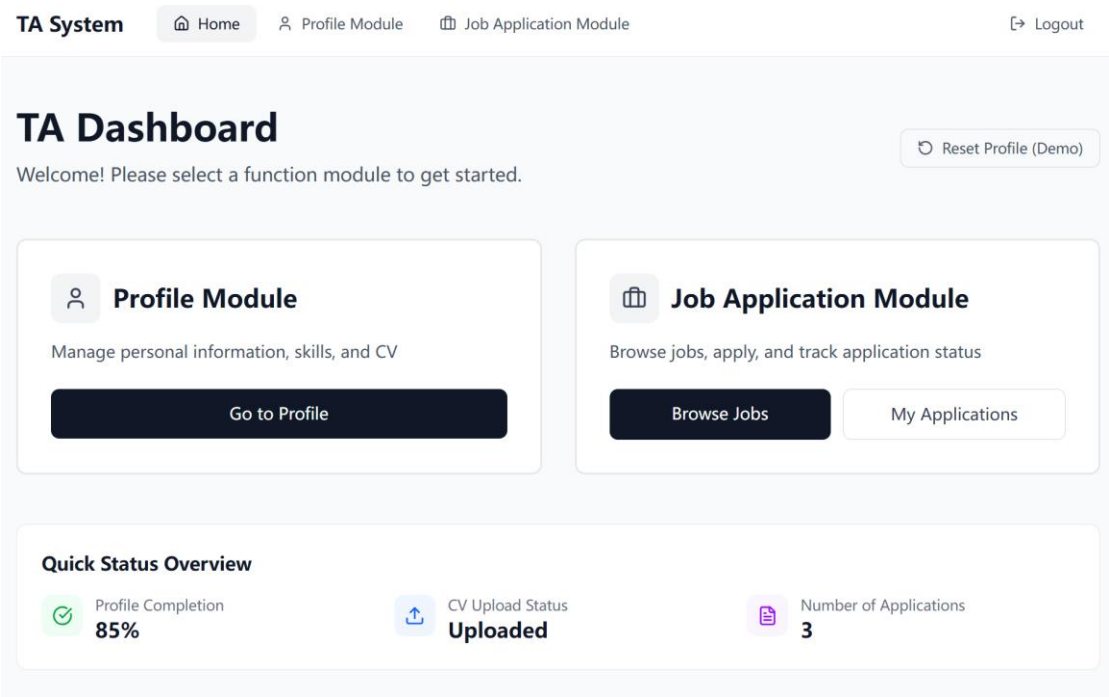


After the user clicks continue on the basic information page, the system enters the "Add Your Skills" page. This page is used to quickly build the user's initial skill profile. On this page, the user can freely input the skills they possess (such as Python, Teaching, Communication, etc.) via a text box and click "Add" to include them. If the user does not wish to fill this out temporarily, they can also click "Skip for now" at the bottom to bypass this step. Once the operation is complete, the user clicks the "Continue to CV Upload" button to transition to the next phase.



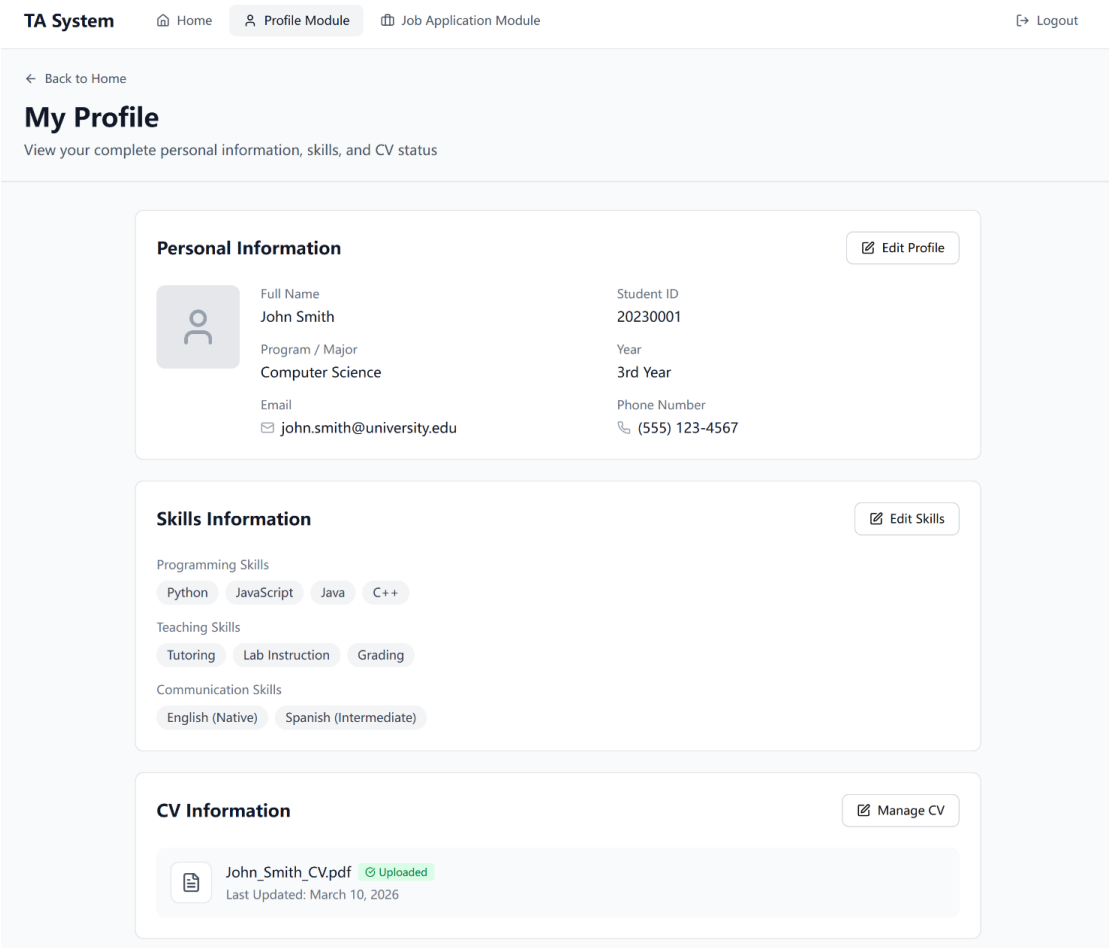
The system then enters the "Upload Your CV" page, which is the final step of the initial profile setup. The center of the page provides an upload area where the user can upload a resume file in PDF, DOC, or DOCX format (maximum limit of 5MB) by dragging and dropping or clicking "Browse File". After the upload is complete (or if "Skip for now" is chosen to bypass), the user clicks the "Complete Profile" button at the bottom right. At this point, the system saves all initial data and formally redirects to the TA system's Dashboard Overview page.

（2） Dashboard Overview



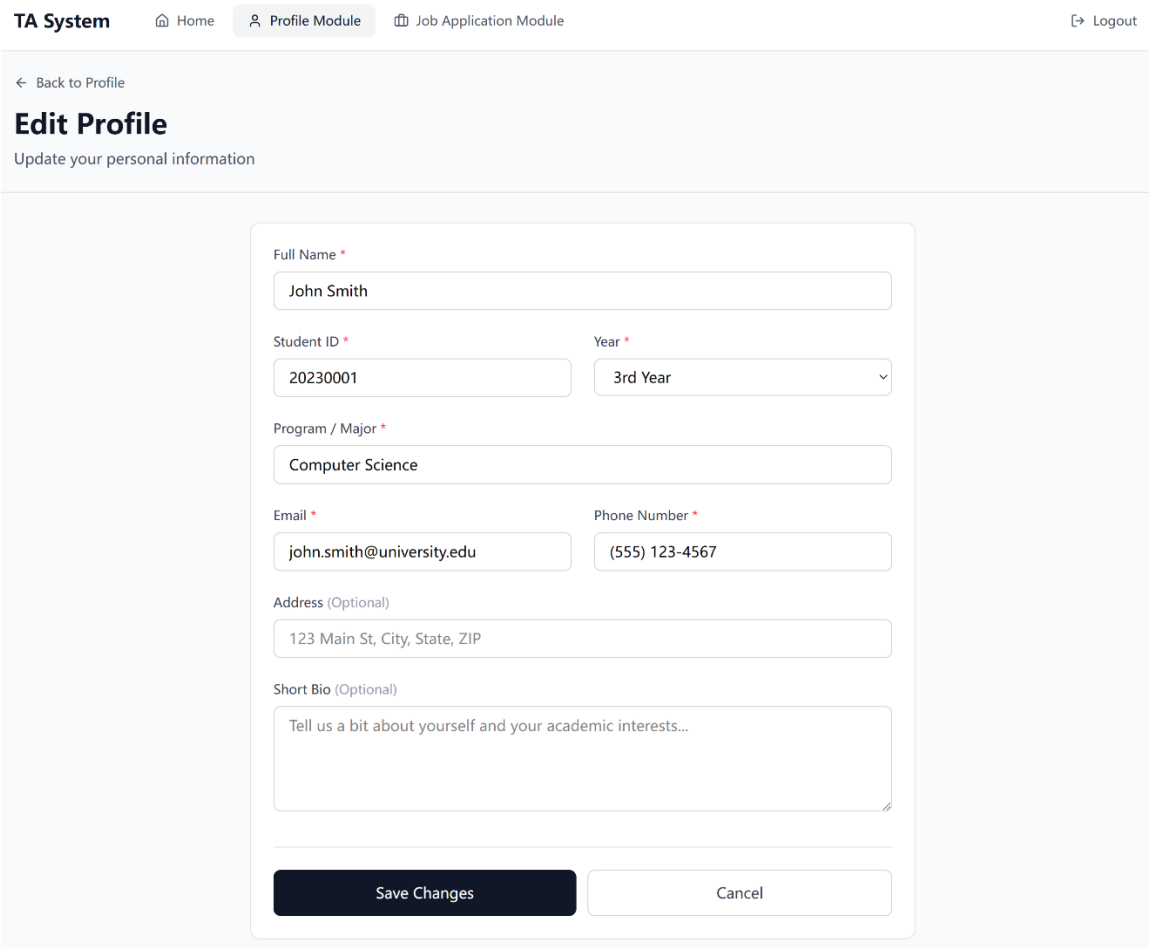
After completing the profile creation, the user first enters the TA Dashboard page. The top navigation bar provides entry points for Home, Profile Module, and Job Application Module, while the main body displays the Profile Module and Job Application Module as two core functional cards. The user can click the "Go to Profile" button within the Profile Module card to directly enter the profile detail page. The "Quick Status Overview" at the bottom of the page displays the Profile Completion (e.g., 85%), CV Upload Status, and the number of applications, providing the user with a quick overview of their readiness before applying for positions.

(3) My Profile



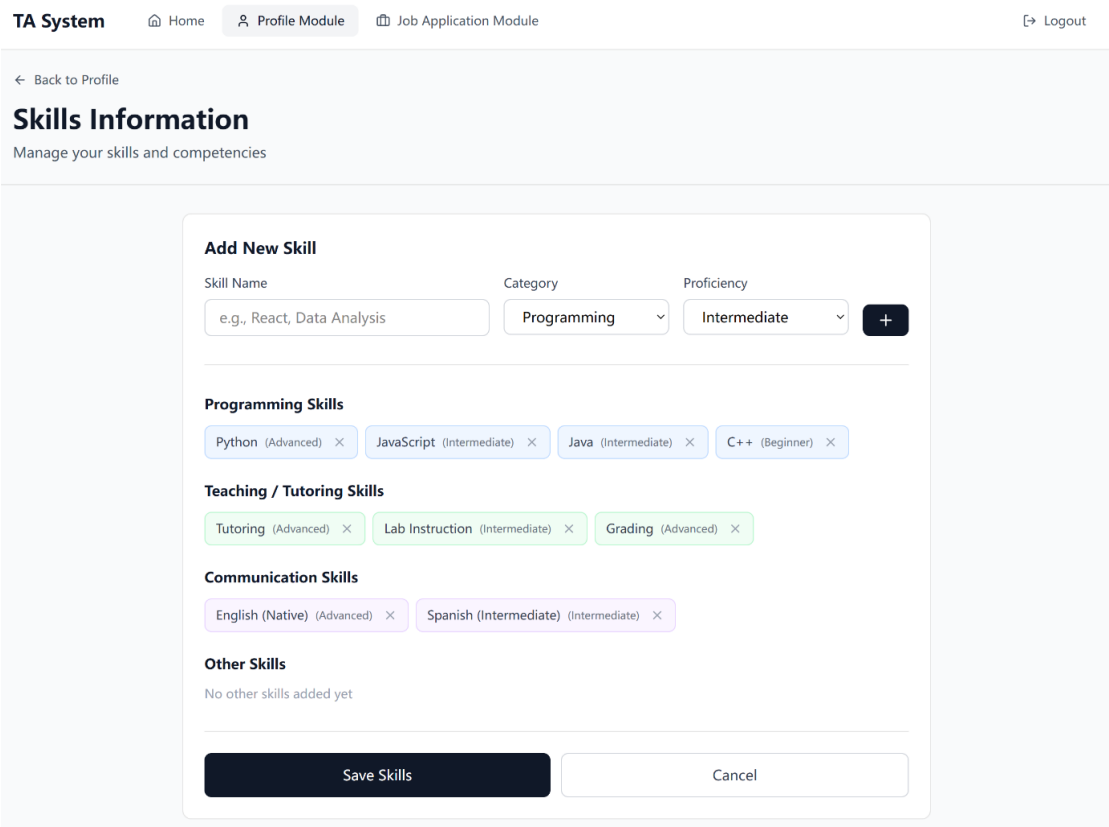
After the user clicks "Go to Profile" on the Dashboard page, the system enters the "My Profile" page. This page displays the user's complete profile information. From top to bottom, the page is divided into three sections: Personal Information, Skills Information, and CV Information, allowing the user to comprehensively review all current attributes and the status of uploaded files. If the user finds that information needs to be supplemented or updated, they can click the "Edit Profile", "Edit Skills", or "Manage CV" buttons located at the top right of the corresponding sections to enter the specific modification workflows. Therefore, the profile detail page serves the roles of information viewing and workflow branching aggregation.

(4) Edit Profile



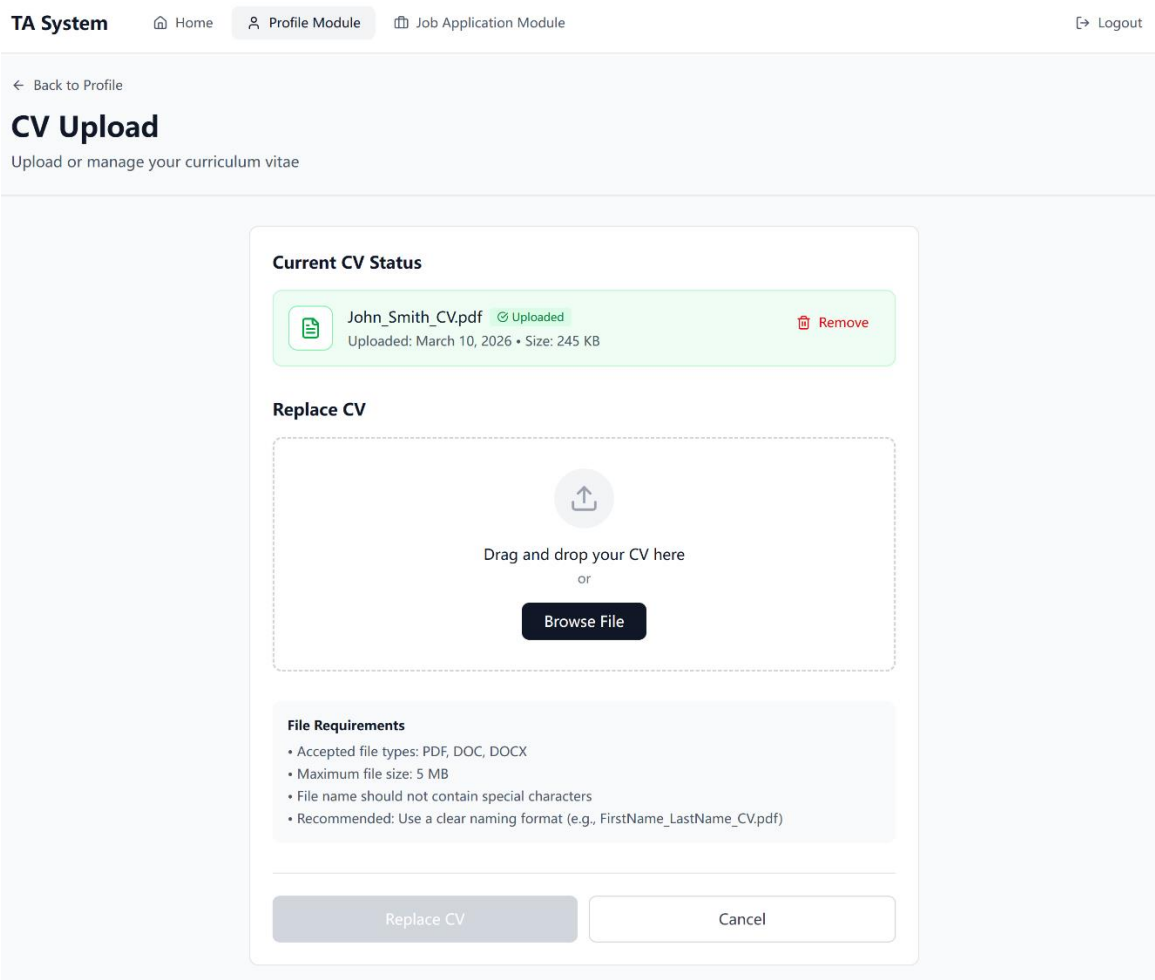
When the user clicks the "Edit Profile" button, the system enters the basic information editing page. This is the core operational page of the profile maintenance workflow. In this form, apart from automatically loading the existing mandatory information, the system also provides additional optional fields, such as Address and Short Bio, to help the user further complete their background description. After filling it out, the user clicks the "Save Changes" button at the bottom, and the system will validate and update the records in the database. If the user decides to abandon the modifications, they can click "Cancel" to leave the current page.

(5) Manage Skills



When the user clicks "Edit Skills" from the detail page, they enter the structured "Skills Information" management page. This page provides a more granular management dimension compared to the initial input. At the top, the user can use dropdown menus to select the Category (e.g., Programming) and Proficiency (e.g., Intermediate) for a new skill. Below this, all saved skills are categorized and displayed as tags; the user can visually review them and click the "X" next to a tag at any time to remove inapplicable items. Once modifications are complete, clicking "Save Skills" synchronizes the updates.

(6) Manage CV

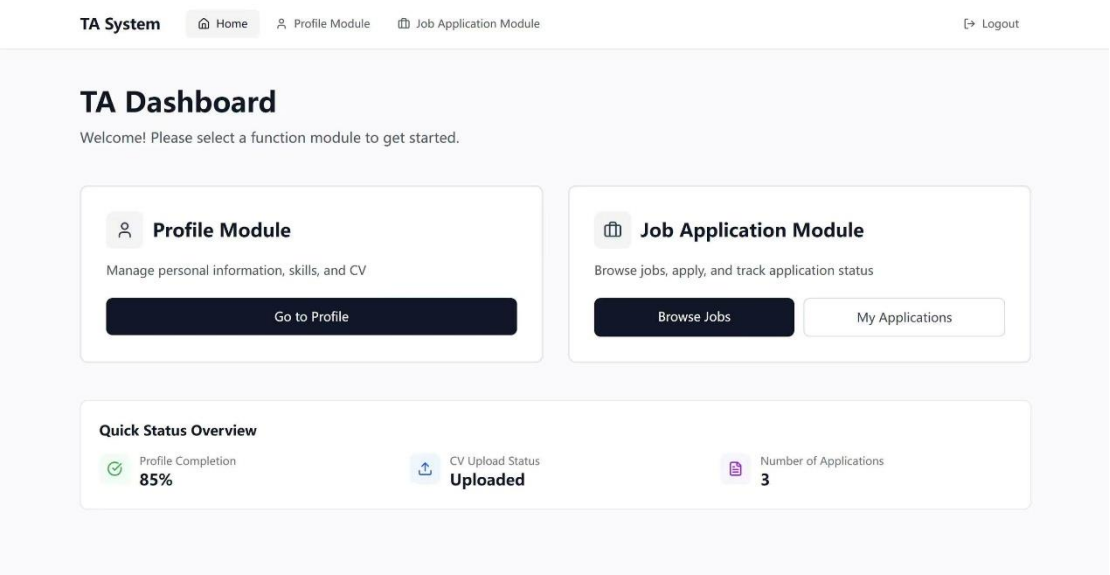


When the user clicks "Manage CV", the system enters the resume version management page. The "Current CV Status" at the top of the page clearly displays the name, upload time, and size of the currently saved file, and provides a one-click "Remove" function. In the "Replace CV" area below, the user can upload a new resume file to overwrite the old version, ensuring that the latest resume is always invoked in the subsequent job application module. After confirming it is correct, clicking the "Replace CV" button saves the changes.

At this point, the TA-side profile module completes a full closed loop from initial creation and status overview to detailed maintenance.

TA Job Application module

(1) Dashboard

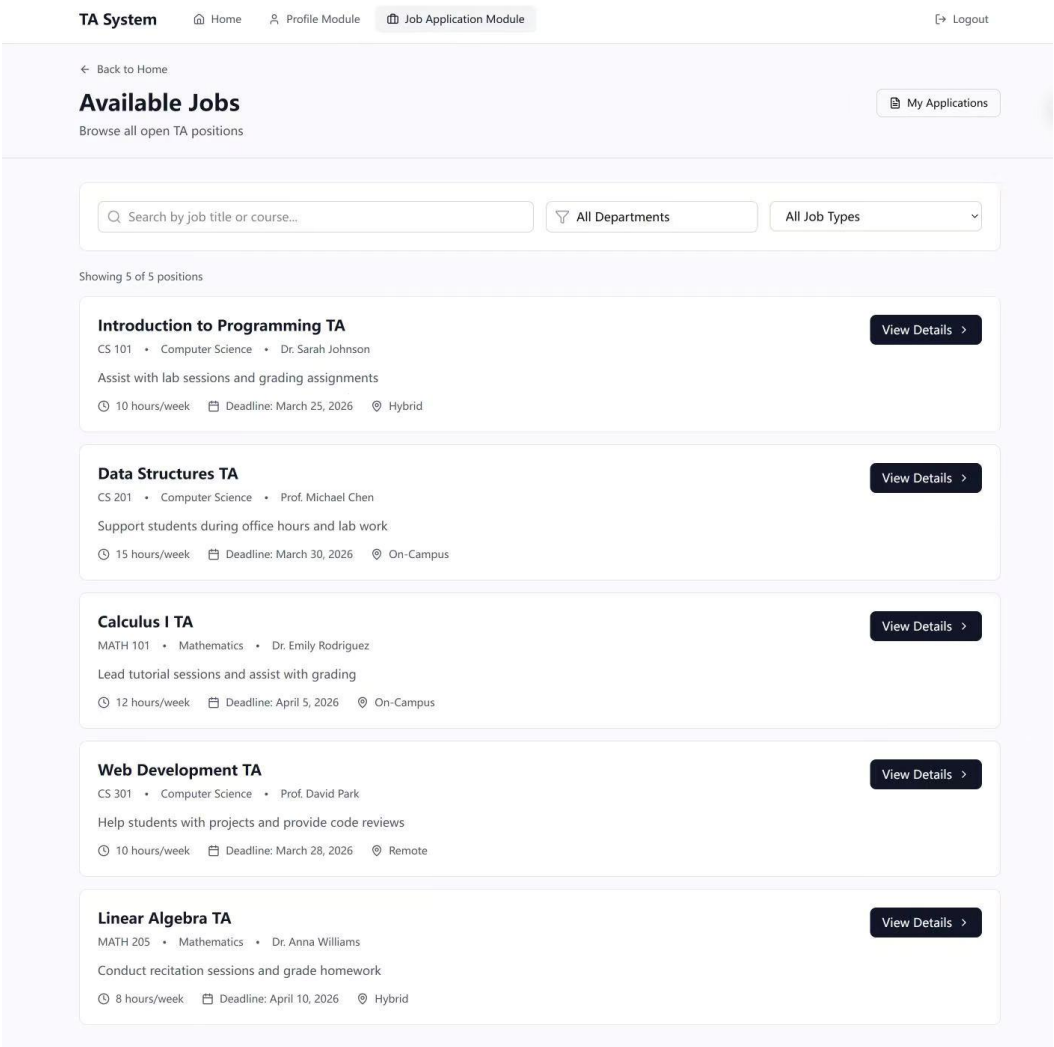


The TA-side job application workflow begins at the system homepage. After logging in, the user first enters the Dashboard page. The top navigation bar provides three main entries: Home, Profile Module, and Job Application Module, while the main body displays Profile Module and Job Application Module as two core functional cards.

In the context of job application, the user may click the Browse Jobs button in the Job Application Module card to enter the job list page and start browsing currently open TA positions. Alternatively, the user may click My Applications to directly review previously submitted applications. Therefore, the dashboard serves both as the starting point of the workflow and as the unified entrance to job browsing and application management.

At the bottom of the page, the Quick Status Overview shows profile completion, CV upload status, and the number of submitted applications, offering users a quick overview of their readiness before applying. If the user finds that personal information is incomplete, they may first go to the Profile Module, complete the necessary information, and then return to the dashboard to continue the application process.

(2) Available Jobs



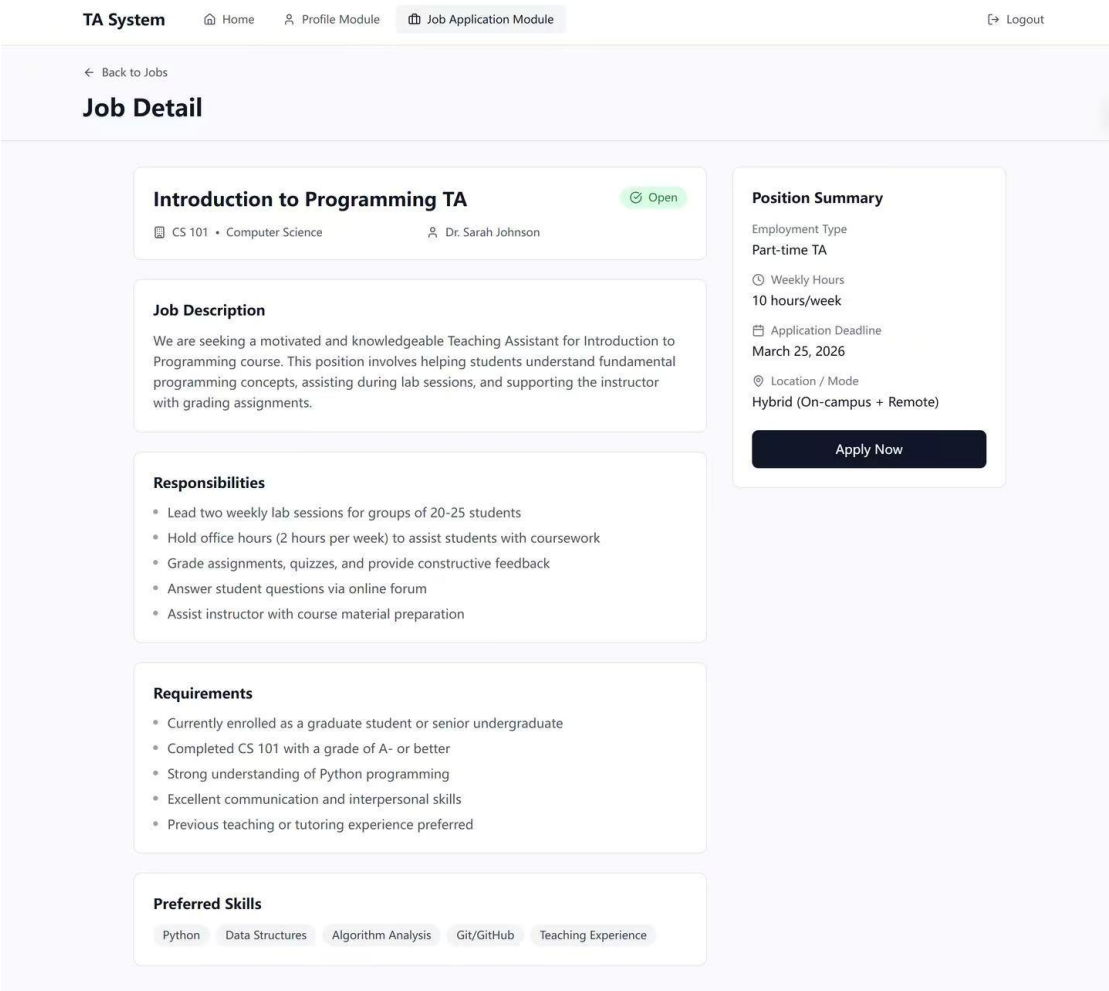
When the user clicks the Browse Jobs button on the Dashboard page, the system enters the Available Jobs page. This page displays all currently open TA positions and functions as the core page for job discovery and filtering.

On this page, the user can enter job titles or course names in the search bar, and may further narrow down the results by department filter and job type filter. The center of the page presents each position as a card, showing summary information such as job title, course code, department, instructor, weekly hours, application deadline, and work mode, which supports preliminary comparison among positions.

Once the user finds a position of interest, they can click the View Details button on the right side of the corresponding job card to enter the job detail page. If the user instead wishes to review previously submitted applications, they may click the My Applications

button at the top right corner to jump directly to the application management page. Therefore, the job list page serves the role of job browsing, filtering, and workflow branching.

(3) Job Detail



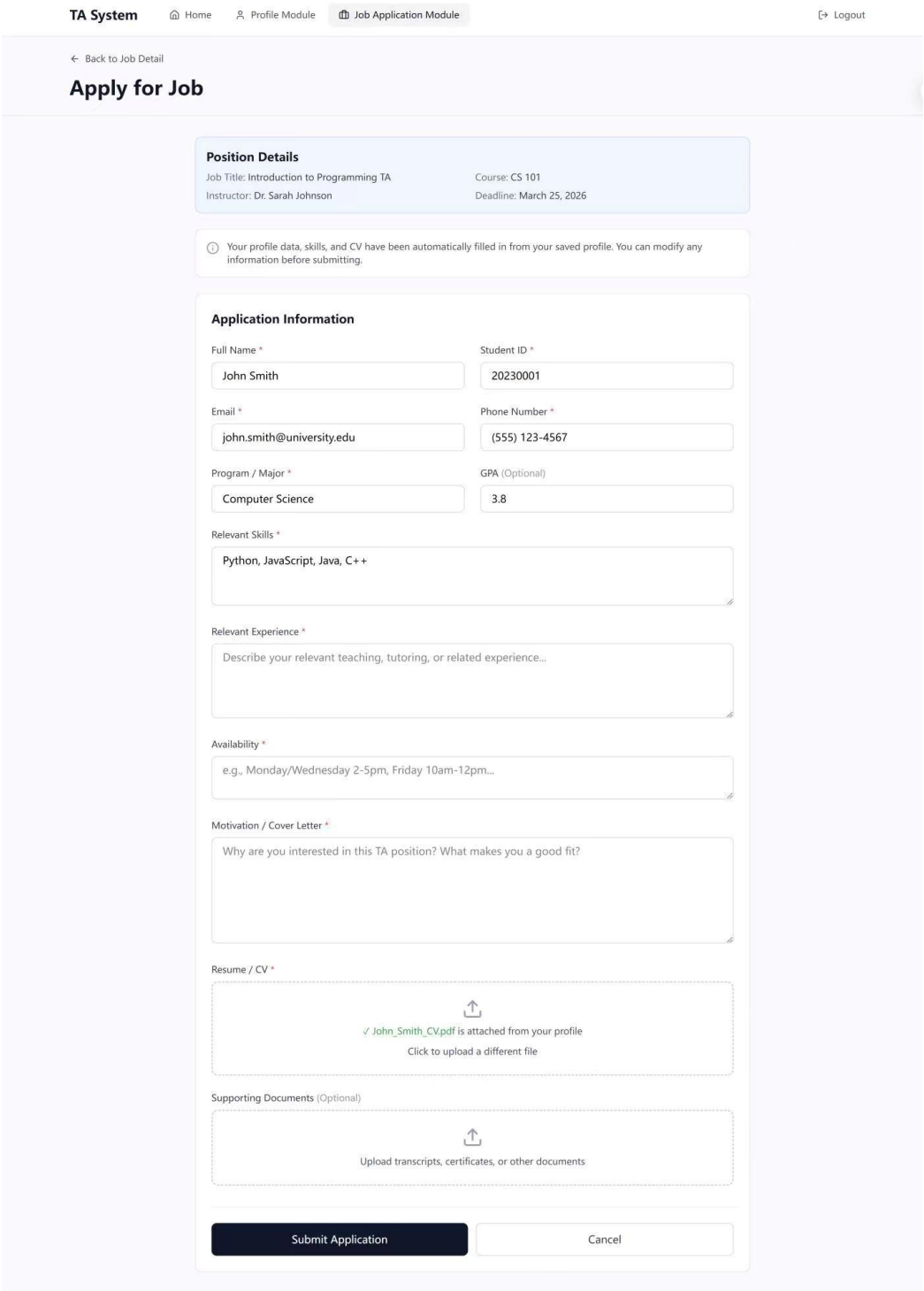
When the user clicks the View Details button for a specific job on the Available Jobs page, the system enters the Job Detail page. This page presents the full information of a specific position and acts as the key decision point where the user determines whether to apply.

On the left side, the page provides detailed information including the job title, course code, department, instructor, and the sections Job Description, Responsibilities, Requirements, and Preferred Skills, allowing the user to fully understand the work

content, eligibility requirements, and preferred competencies. On the right side, the Position Summary highlights important information such as employment type, weekly hours, application deadline, and work mode.

After reviewing the job information, if the user believes that their background matches the requirements, they can click the Apply Now button on the right to proceed to the application form page. If the user decides that the position is not suitable, they can click Back to Jobs at the top left corner to return to the job list page and continue browsing other opportunities. Thus, the job detail page serves as the crucial transition point from job exploration to formal application.

(4) Apply For Job



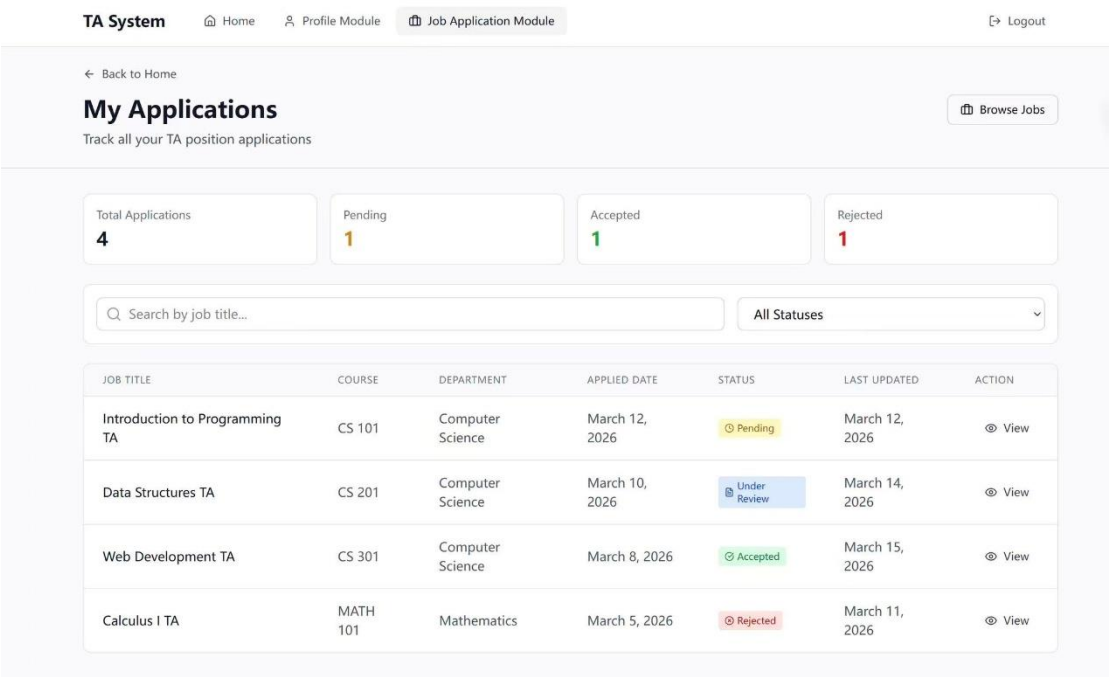
When the user clicks the Apply Now button on the Job Detail page, the system enters the Apply for Job page. This is the core operational page of the application workflow, where the user fills in and submits the formal application information.

At the top of the page, the Position Details section displays the job title, course, instructor, and deadline again, helping the user confirm the target position. In the form

area, part of the basic information has already been automatically filled in from the Profile Module, so the user only needs to verify the content and supplement position-related information such as relevant skills, experience, availability, and motivation. In addition, the page supports uploading a CV and optional supporting documents.

After completing the form, the user clicks the Submit Application button at the bottom of the page. The system then validates the information, saves the application record, and initializes the application status. If the user does not wish to continue, they may click Cancel to abandon the current operation and leave the page. This page completes the most critical step in the workflow, transforming job interest into a formal submission.

(5) My Applications



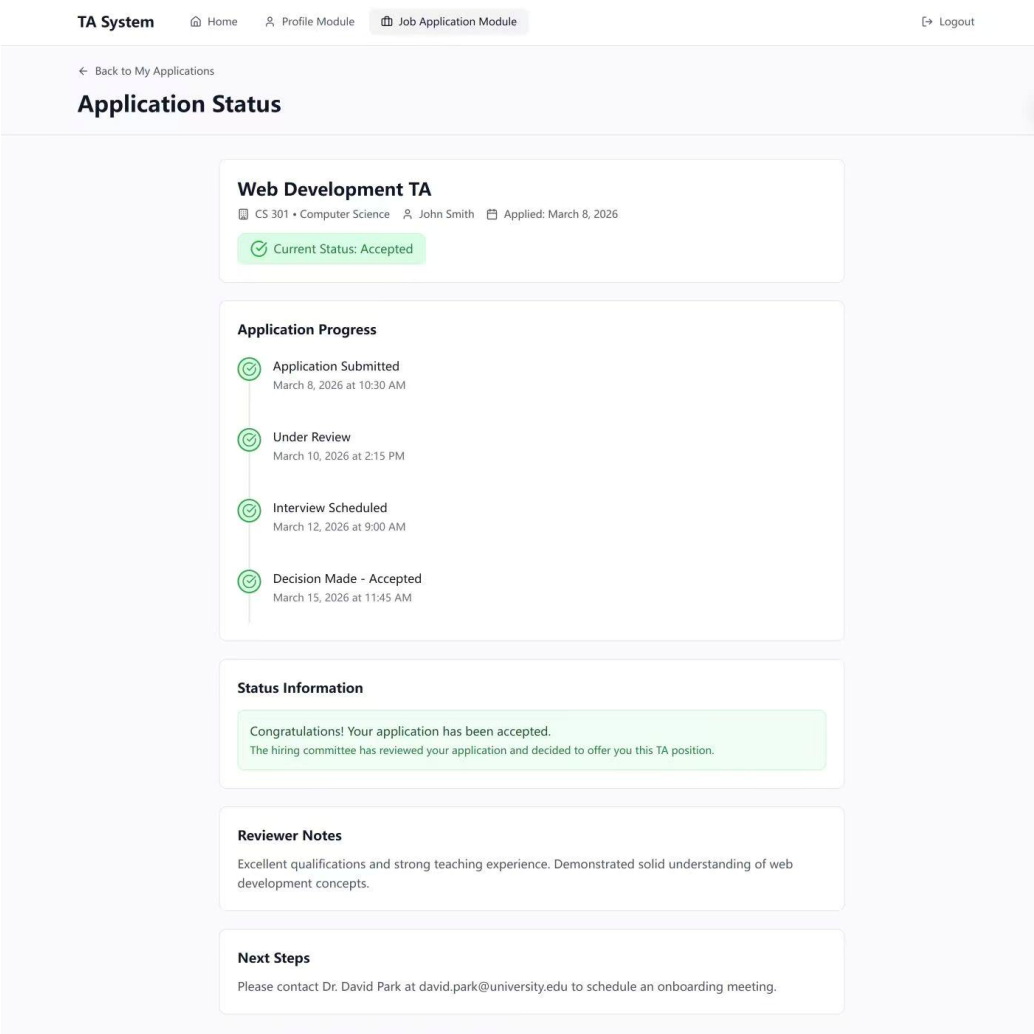
After the user successfully clicks Submit Application, the system creates a new application record and includes it in the My Applications page for centralized management. The user may also actively enter this page by clicking My Applications from either the Dashboard or the job list page.

At the top, summary cards display the total number of applications and the number under different statuses such as Pending, Accepted, and Rejected, helping the user

quickly understand the overall application situation. The table below lists each application's job title, course, department, applied date, current status, and last updated time, allowing the user to clearly review the progress of every submission.

If the user wants to check the detailed processing history of a specific application, they can click the View button on the far right of the corresponding row to enter the application status detail page. Therefore, the My Applications page serves to receive the result of submission, centralize application management, and connect to the next stage of status tracking.

(6) Application Status



When the user clicks the View button on the right side of a specific application record on the My Applications page, the system enters the Application Status page. This page displays the full processing history of a single application and serves as the final key

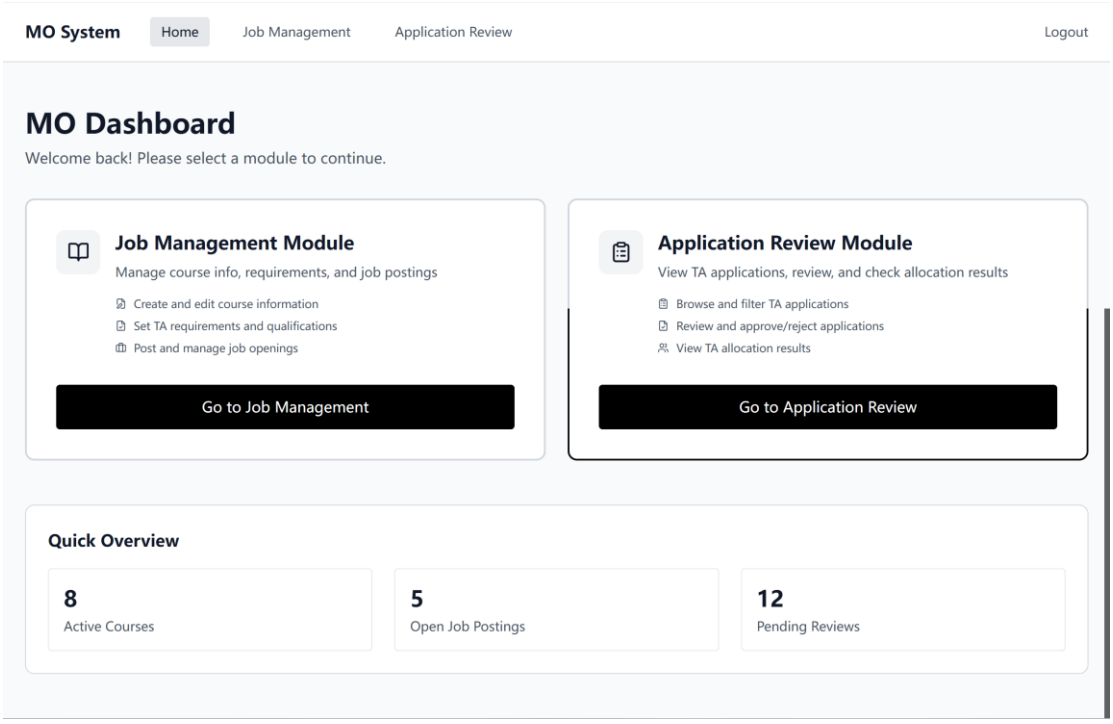
page in the workflow.

At the top, the page shows the job title, course code, applicant, application date, and current status, allowing the user to quickly recognize the current outcome. In the middle, the Application Progress section visualizes the full process from Application Submitted to Under Review, Interview Scheduled, and finally the decision result, thereby improving transparency.

At the bottom of the page, the system further uses Status Information to provide the formal decision, Reviewer Notes to show evaluation comments, and Next Steps to guide the user on follow-up actions. For example, when the application result is Accepted, the page instructs the user to contact the relevant instructor for onboarding arrangements. At this point, the TA-side job application workflow completes a full closed loop from job browsing and application submission to result feedback.

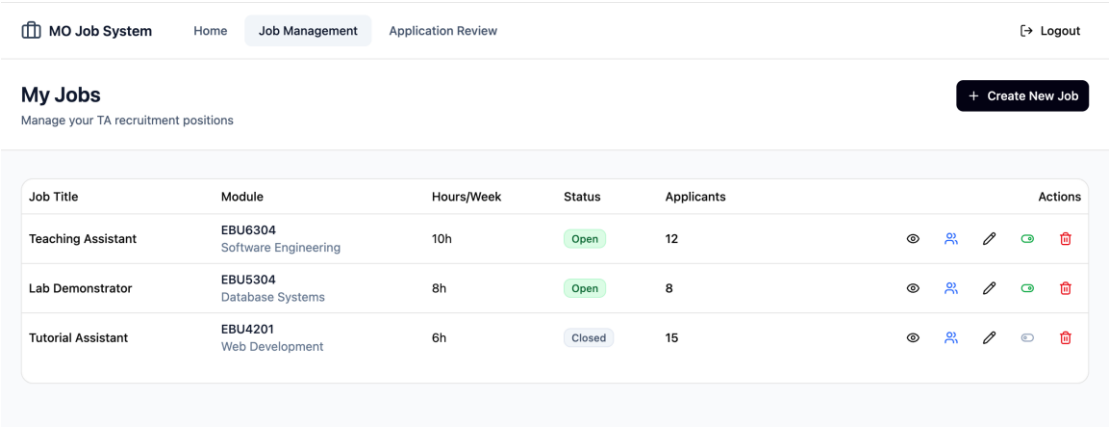
MO Job Management Module

(1) MO Dashboard



Upon logging into the system, the MO first enters the MO Dashboard, which serves as the central command center for all TA recruitment activities. The top navigation bar and the main content area cleanly separate the system into two distinct workflows: the 'Job Management Module' on the left and the 'Application Review Module' on the right. Each module card clearly outlines its core functions using a list and features a prominent primary button (e.g., 'Go to Job Management') to efficiently guide the MO to their target operational page. Furthermore, the 'Quick Overview' section at the bottom of the page displays real-time key operational metrics—such as 8 Active Courses, 5 Open Job Postings, and 12 Pending Reviews.

(2) My Jobs



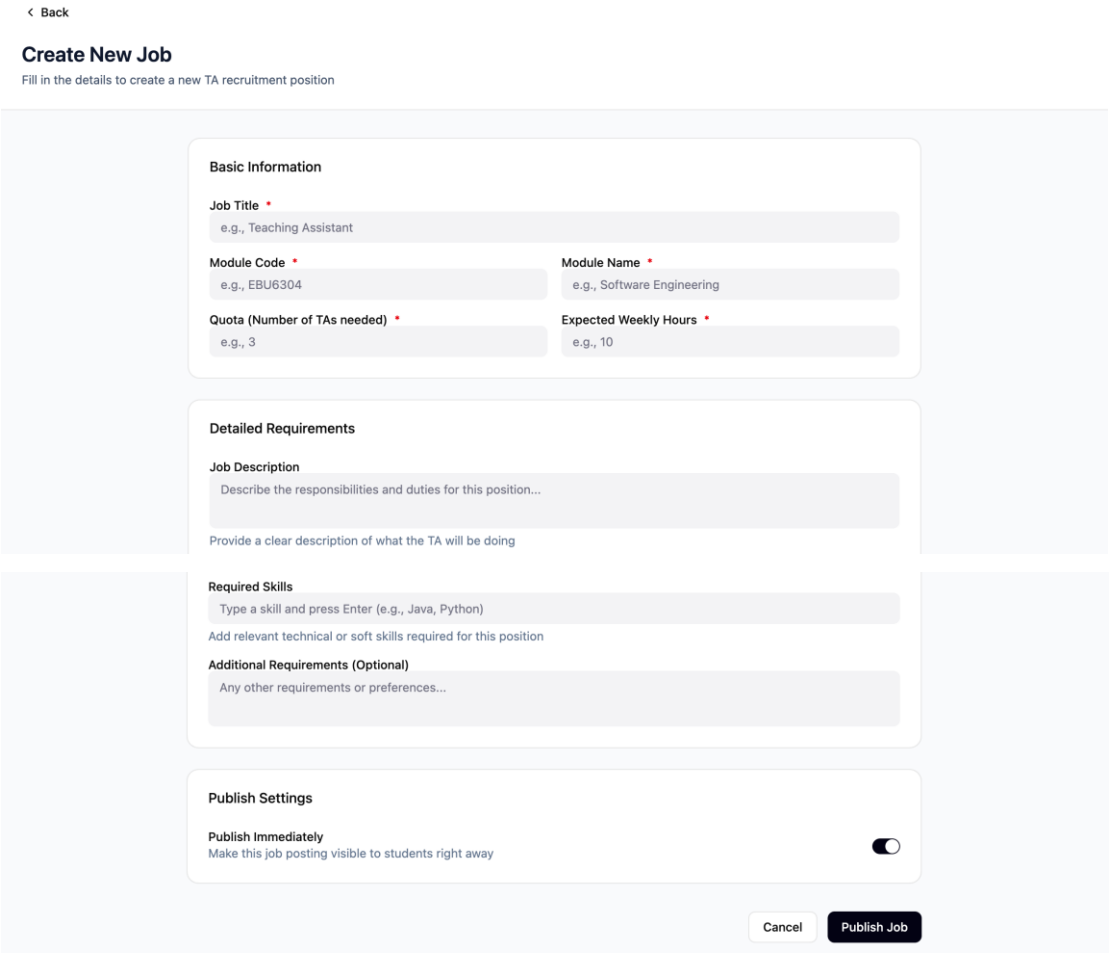
The "My Jobs" page serves as the centralized control panel for all TA recruitment positions within the system. To initiate a new recruitment cycle, the Module Organiser (MO) can click the prominent "+ Create New Job" button located at the top right.

The main interface features a clean, structured data table that displays key metrics for each posting: Job Title, Module, Hours/Week, current Status (such as Open or Closed), and the total number of Applicants.

For efficient job lifecycle management, the far-right "Actions" column provides a suite of five quick-access icons for each row. From left to right, these icons allow the MO to instantly perform the following key actions:

- **View Job Details (Eye icon):** Review the full description and requirements of the posting.
- **View Allocated TAs (User Group icon):** Instantly access the contact list of hired candidates for that specific module.
- **Edit Job (Pencil icon):** Modify the job posting details and parameters.
- **Toggle Status (Switch icon):** Seamlessly change the job's visibility status between open and closed.
- **Delete Job (Red Trash Bin icon):** Permanently remove the recruitment record from the system.

(3) My Jobs

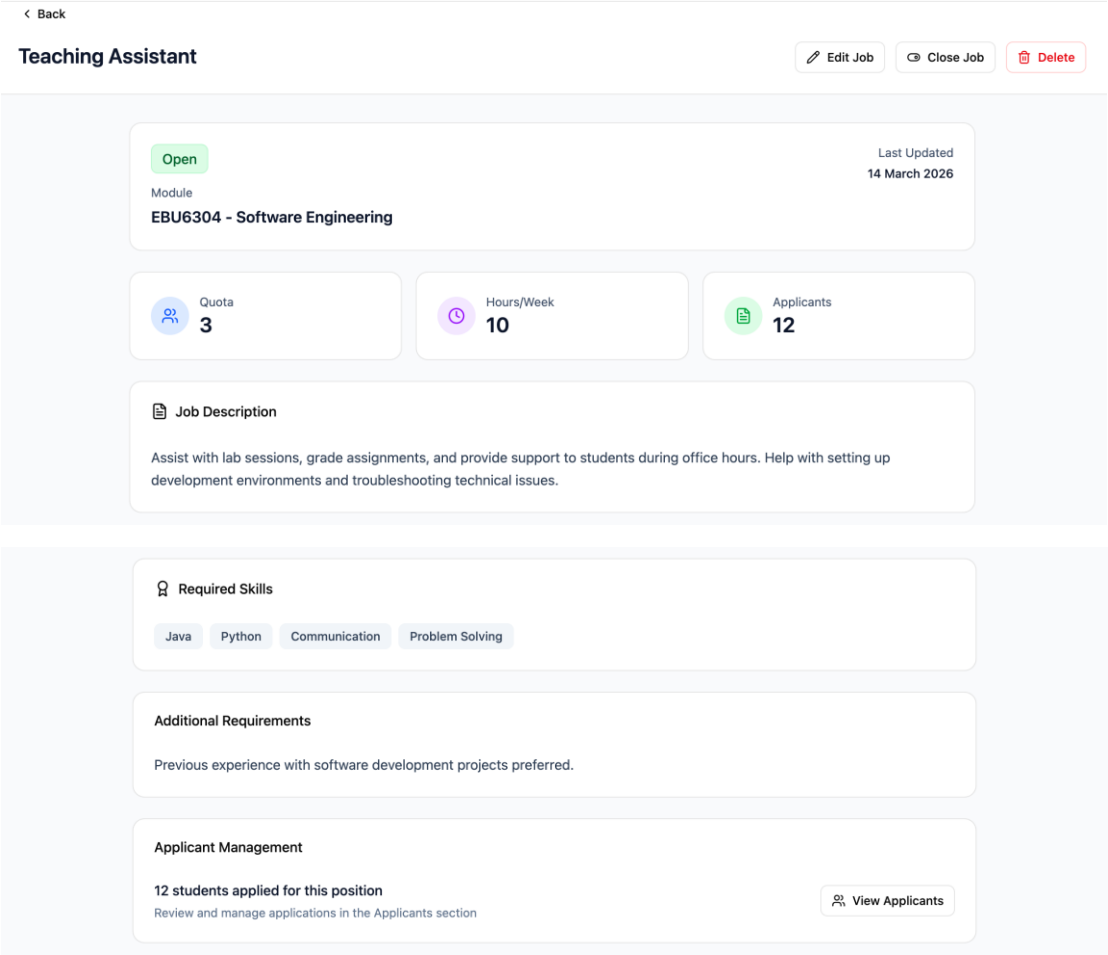


The "Create New Job" page is the primary interface for publishing recruitment needs. To ensure data quality, the form is divided into three distinct sections:

- Basic Information: This section captures mandatory fields (marked with an asterisk). The MO must input the Job Title, Module Code, Module Name, Quota, and Expected Weekly Hours. This structured data is crucial for filtering on the student side.
- Detailed Requirements: Here, the MO provides the qualitative context. It includes a free-text field for the Job Description and a "Required Skills" input where typing a skill and hitting Enter generates a distinct tag. An optional field is also available for additional preferences.
- Publish Settings: A simple "Publish Immediately" toggle allows the MO to control visibility. Turning it on pushes the job live to the TA system; turning it off saves it as a draft.

Clicking "Publish Job" validates the form, commits the new record to the backend system, and redirects the MO back to the Job Management dashboard.

(4)Job Detail

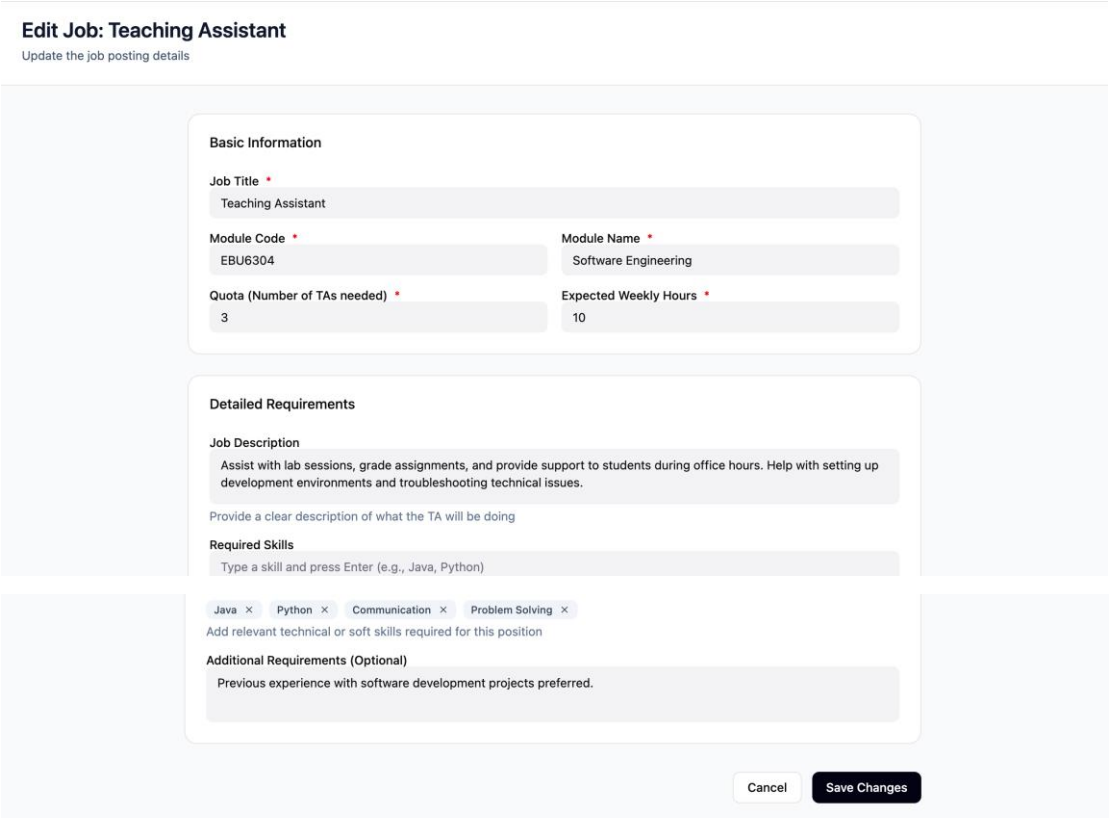


The MO accesses the "Job Detail" page by clicking the "Eye" icon on a job record in the "My Jobs" list. This page displays the complete information of a specific job posting in a read-only format.

The top section shows the Job Title and provides action buttons to Edit Job, Close Job, and Delete the posting. Directly below, data cards display key metrics: the current Status, Module details, Quota, Hours/Week, and the number of Applicants.

The middle section lists the Job Description, Required Skills (displayed as tags such as Java and Python), and Additional Requirements. At the bottom of the page, the "Applicant Management" section shows the total number of applications received and includes a "View Applicants" button. Clicking this button redirects the user to the Application Review module.

(5)Edit Job

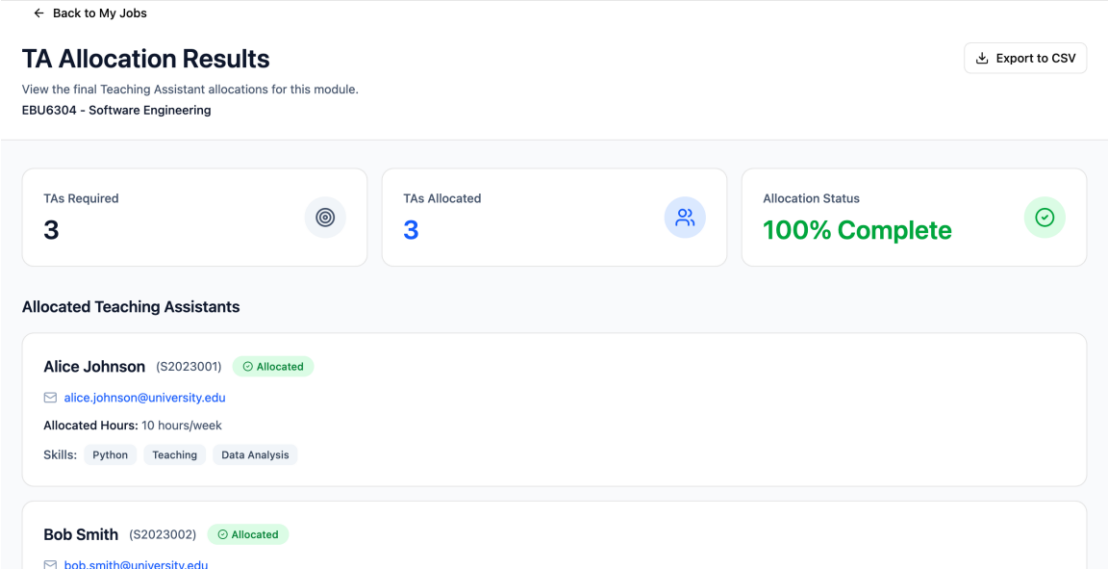


The MO accesses the "Edit Job" page by clicking the pencil icon in the job list or the edit button on the job detail page. This page allows users to update the details of an existing job posting. The form layout is identical to the creation page, but all input fields are pre-filled with the current data of the selected job.

In the "Basic Information" section, the MO can modify mandatory fields such as the Job Title, Module Code, Quota, and Expected Weekly Hours. The "Detailed Requirements" section allows updates to the Job Description, Required Skills, and Additional Requirements. The Required Skills are displayed as tags with an "x" icon, allowing users to directly remove existing skills or add new ones.

At the bottom of the page, the user can click the "Save Changes" button to apply the updates to the system, or click the "Cancel" button to discard any modifications and return to the previous screen.

(6) TA Allocation Results



The MO accesses the "TA Allocation Results" page by clicking the "People" icon on a job record in the "My Jobs" list. This page displays the final list of hired TAs for a specific module.

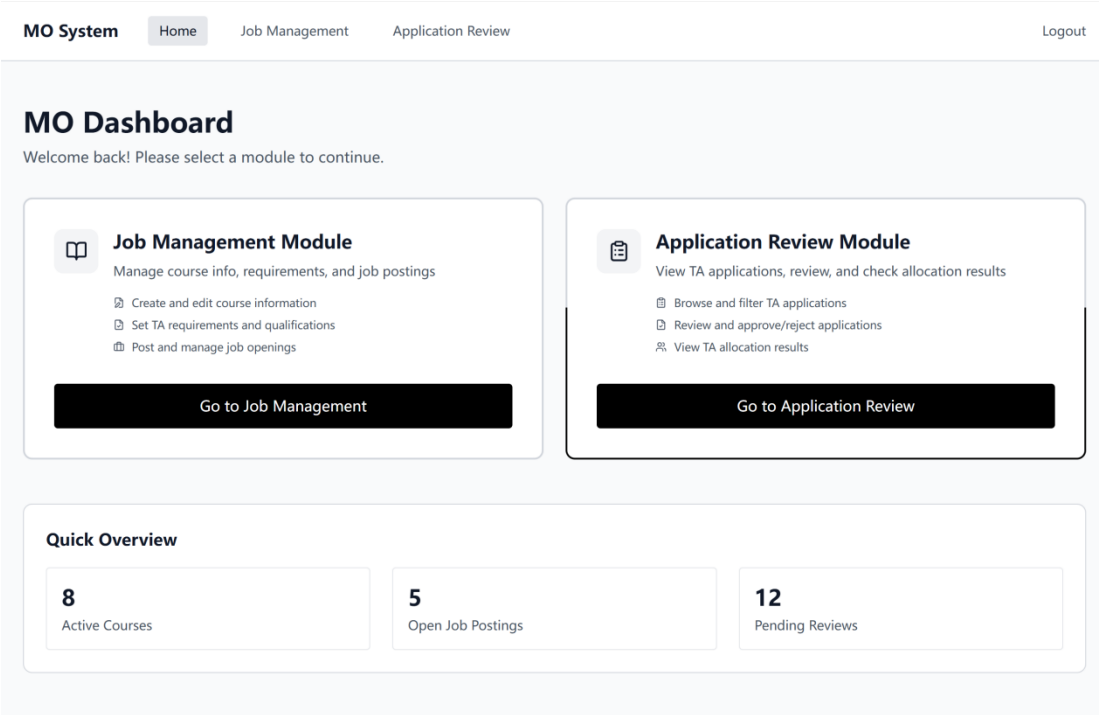
The header shows the module details and includes an "Export to CSV" button at the top right. Below the header, three data cards summarize the recruitment progress: TAs Required, TAs Allocated, and the overall Allocation Status (e.g., 100% Complete).

The main section lists the Allocated Teaching Assistants. Each record provides the TA's name, student ID, an "Allocated" status badge, contact email, allocated working hours, and relevant skill tags (e.g., Python, Teaching).

Application Review Module

This system is an assistant recruitment management platform designed for Module Organizers (MO), featuring two core modules: Course Management and Application Review. It supports end-to-end operations including course information maintenance, TA position postings, application review, and result tracking. With a mid-fidelity web interface design, the platform offers intuitive and user-friendly operations, facilitating modular development and team collaboration.

（1）MO Dashboard

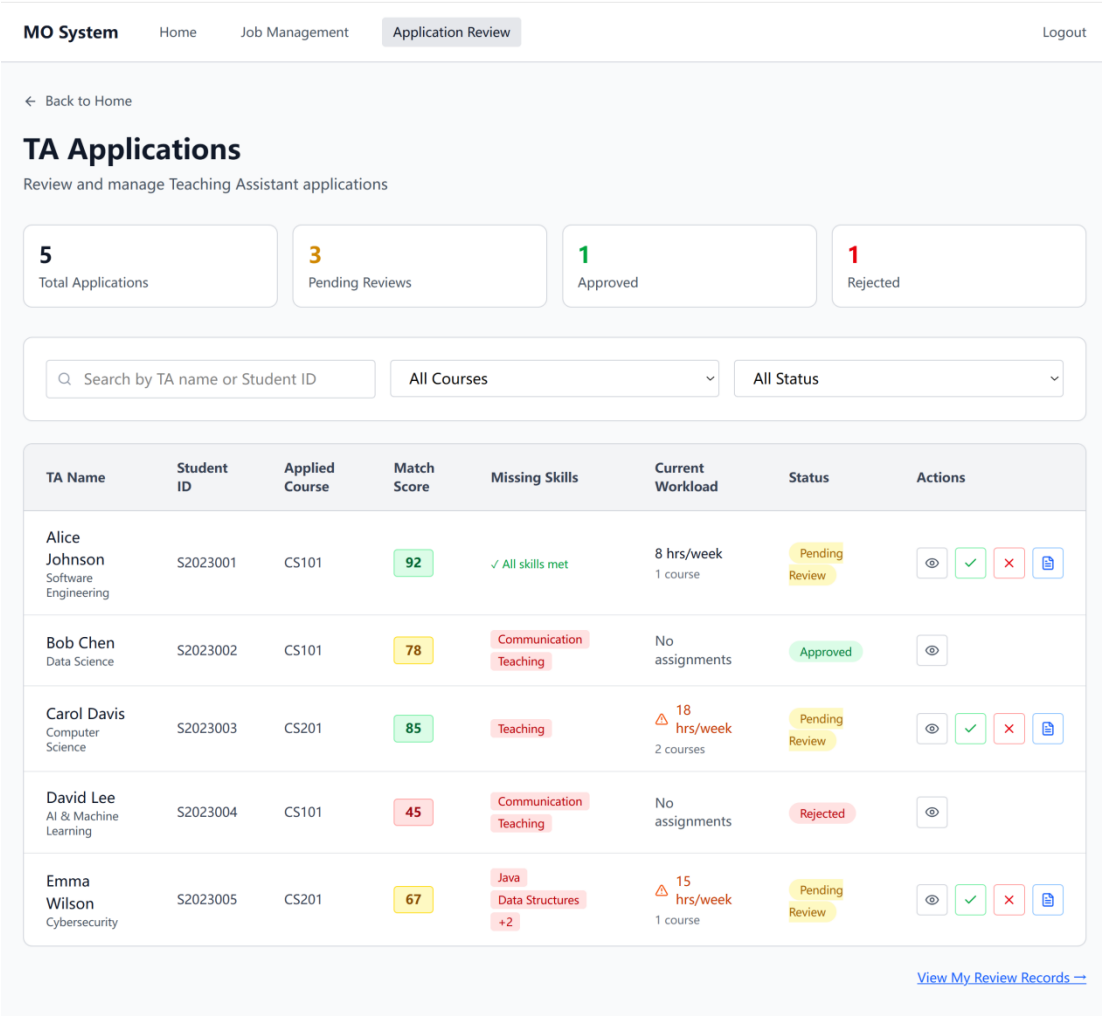


The homepage is the default page after login, featuring core functions such as module entry navigation and data overview. The module entry includes Job Management module and Application Review module.

- Job Management Module: Manage course information, TA requirements, and job postings. Click "Go to Job Management" to access it.
- Application Review module: Browse TA applications, complete reviews, and view assignment results. Click "Go to Application Review" to enter.
- Quick Overview: Active Courses, Open Job Postings, Pending Reviews.

When you first log in, you can quickly assess your workload through these overview metrics before selecting relevant modules to begin your tasks.

（2）TA Applications



Clicking "Go to Application Review" will take you to the TA Applications page, which offers core features including browsing, filtering, and fast processing of TA applications.

- Data statistics: displays the total number of applications, pending approvals, approved approvals, and rejected approvals.
- Screening and Search: Supports searching by TA name/student ID; searching by course; filtering by application status (All/Under Review/Approved/Rejected).
- The application list includes the following details:

  TA Name, Student ID, Applied Courses;

  Match Score: Displays a skill matching score ranging from 1 to 100, coded with

different colors (green ≥ 80, yellow ≥ 60, red < 60);

Missing Skills: Highlights key skills lacking by applicants using red labels, "✓ All skills met" in green if all requirements are satisfied;

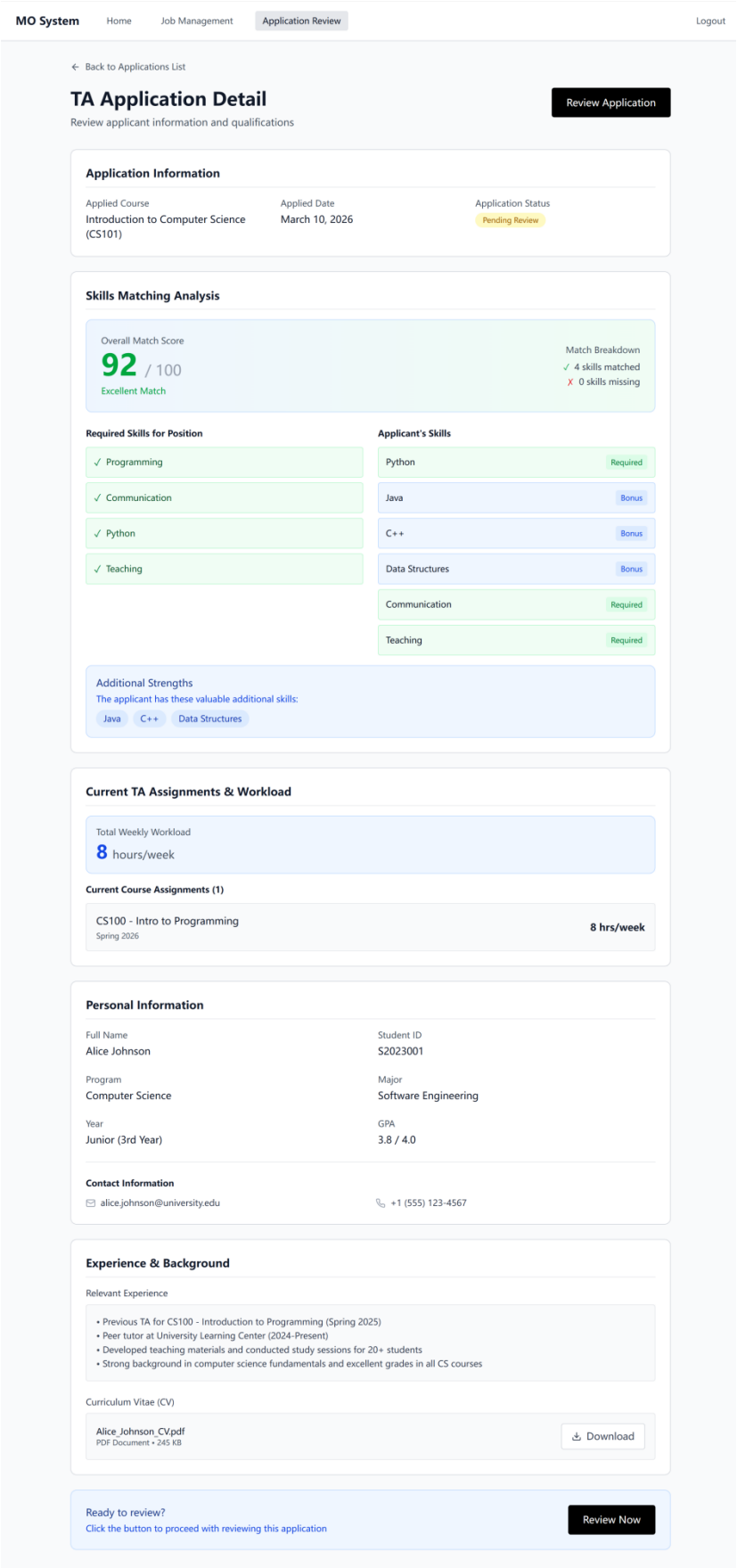
Current Workload: Shows weekly workload hours per applicant (avoiding scheduling conflicts with other modules), where 0 hours indicate "No assignments" (gray), 1-14 hours represent normal workload, and ≥15 hours display an orange warning icon with workload alerts for high-risk scenarios. Below workload shows assigned course count;

Status: Pending review / Approved / Rejected;

Quick action buttons include: View application details, Quick approve (✓), Quick reject (✗), and Full application review.

- In the lower right corner of this page, you can click "View My Review Records→" to directly view historical review records and statistics.

（3）TA Application Detail



On the Application List page, click "View Application Details" to access the page, which includes viewing applicant information and skill matching analysis.

- Application Information: Displays the applied course, application date, and current status (Pending Review / Approved / Rejected).
- Skills Matching Analysis:

  Overall match score (out of 100) and match evaluation;

  Position required skills ***VS*** applicant skills comparison: The left side displays the required skills for the position (green ✓= matched, red ✗= missing), while the right side shows the applicant's skills (green = required, blue = additional);

  Red warning box: Highlights missing skills. Blue prompt box: Displays the applicant's additional skill advantages.
- Current TA Assignments & Workload: Display the courses the applicant has completed and their weekly working hours to help determine if they are overloaded or if students have been hired by other modules.

  Total Weekly Workload: Displays total workload in large size, using different colors (gray/blue/orange) based on workload levels;

  Current Course Assignments: Shows all assigned courses for students in detail, including course name, semester, and weekly workload;

  Workload Conflict Warning: Displays an orange warning box when total workload reaches or exceeds 15 hours, indicating potential scheduling conflicts and performance impacts;

  Empty Status Prompt: Shows a friendly message if no courses are currently assigned to the student.
- Personal Information: Displays basic information such as name, student ID, major, GPA, and contact details.
- Experience & Background:

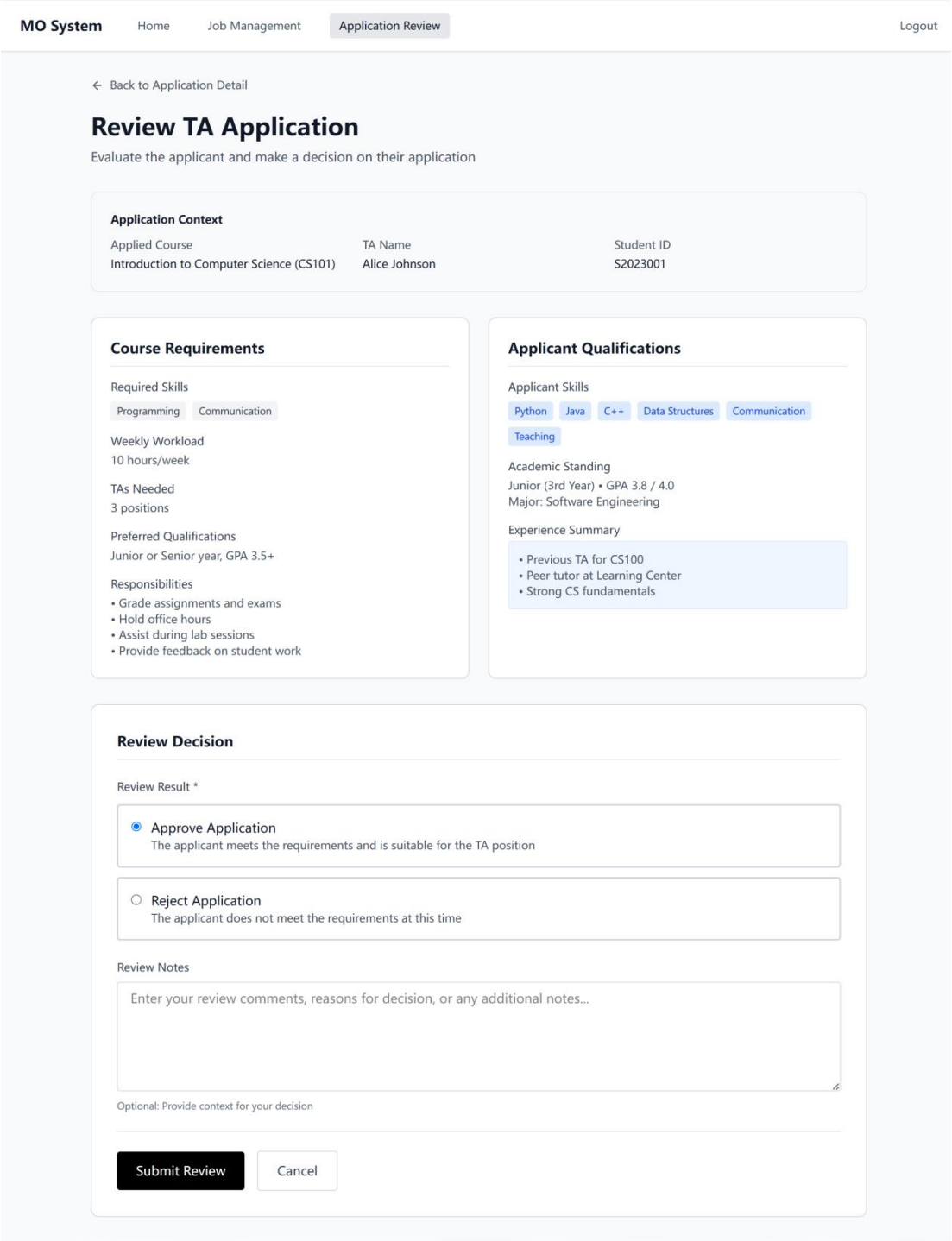
  Description of relevant teaching/teaching assistant experience;

  CV file download portal.

  Meanwhile, click "Review Application" in the upper right corner of the page or

"Review Now" at the bottom to access the review page.

（4）Review TA Application



On the "TA Application Details" page, click "Review Application" in the top-right corner or "Review Now" at the bottom to access the review page. This page allows you to complete the application review and record decisions.

- Application Context: displays the requested course, TA name, and student ID.
- Course Requirements ***VS*** Applicant Qualifications:

  Left side —— Course Requirements: Skills required for the course, weekly workload, number of teaching assistants (TAs) needed, preferred qualifications, and job responsibilities;

  Right side —— Applicant qualifications: Applicant's skills, academic standing, and experience summary.
- Review Decision:

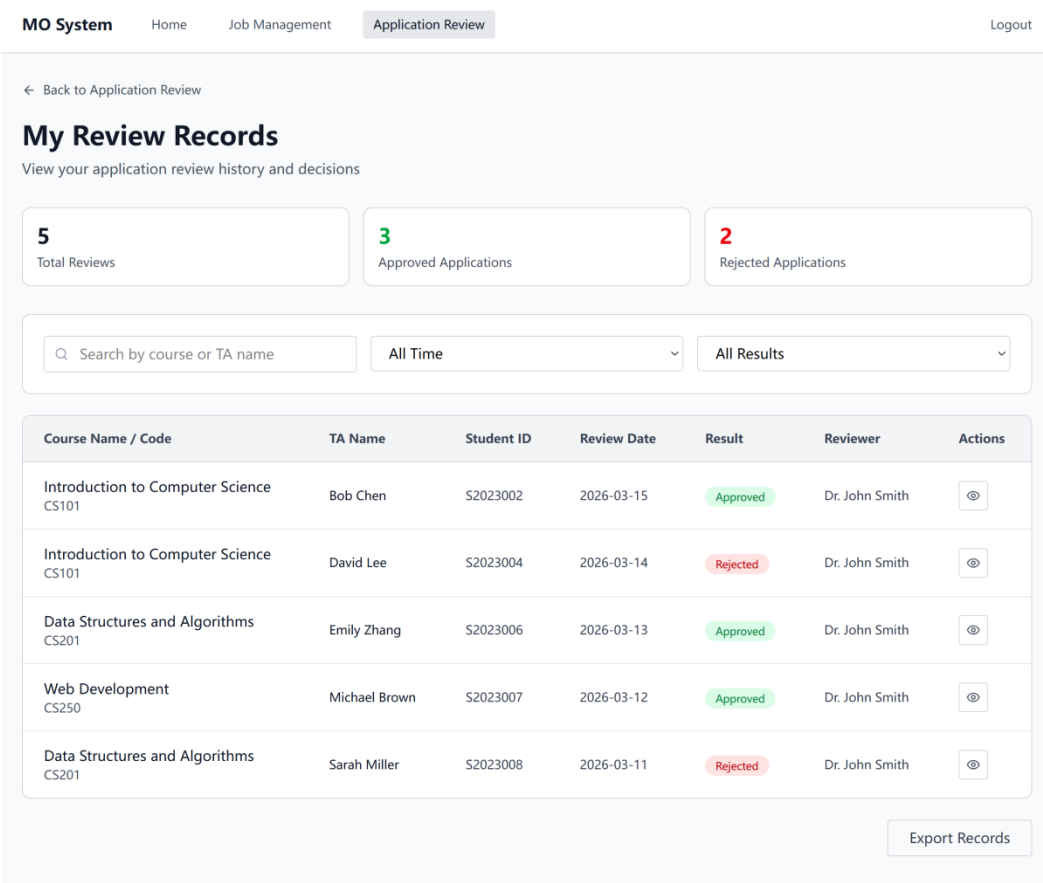
  Select "Approve Application" or "Reject Application";

  Enter the review reason or notes in "Review Notes".
- Submit Operation:

  Click "Submit Review" to complete the review and update the status;

  Click "Cancel" to abandon the current operation and return to the previous page.

（5）My Review Records

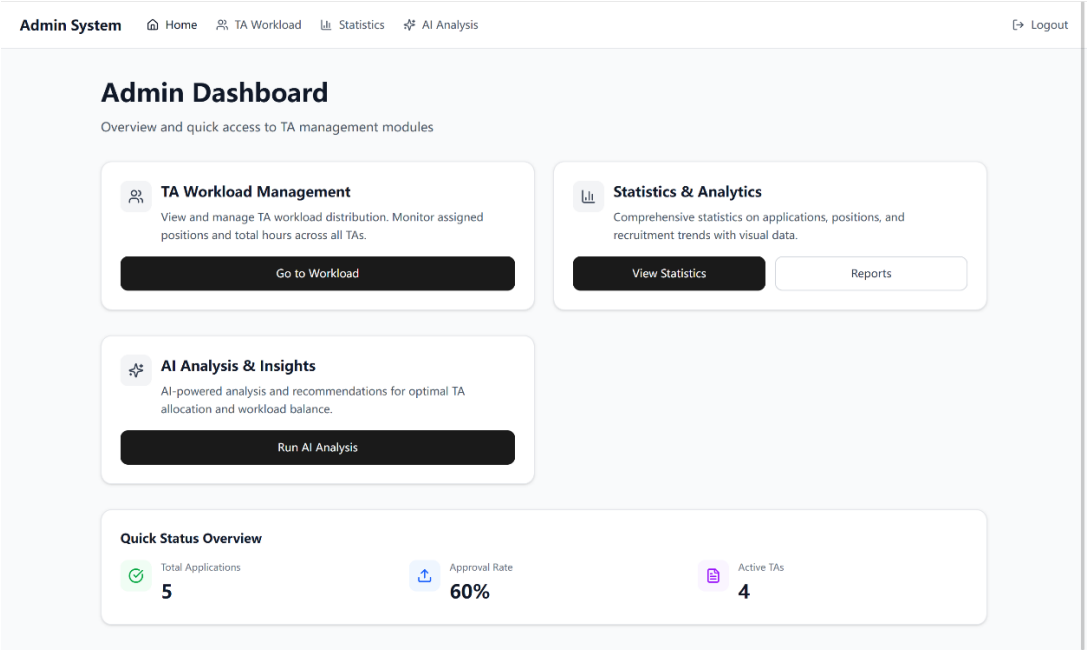


Click "View My Review Records" in the lower right corner of the TA Applications page to access the review records page, where you can view historical review records and statistics.

- Data statistics: displays the total number of reviews, approved numbers, and rejected numbers.
- Filtering and Search: Search by course/TA name, time range, or review status.
- Record list: displays Course Name/Code, TA Name, Student ID, Review Date, Result, and Reviewer.
- Operation: Click "View Application Details" to view the audit details of a single record.
- Export function: Click "Export Records" in the lower right corner of the page to export all audit records as files.

Admin System Management Module

(1) Admin Dashboard



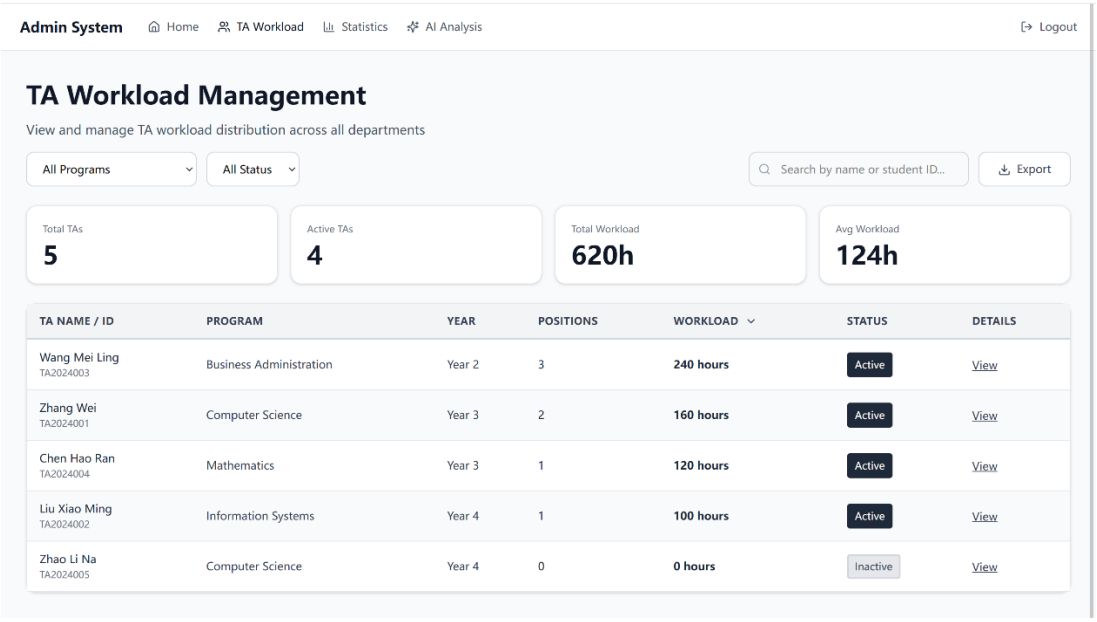
After logging into the system, the administrator is directly directed to the **Admin Dashboard**, the homepage of this module. The system automatically performs three key tasks:

It fetches all basic data from the database in one go, including job postings, TA application records, and TA personal profiles.

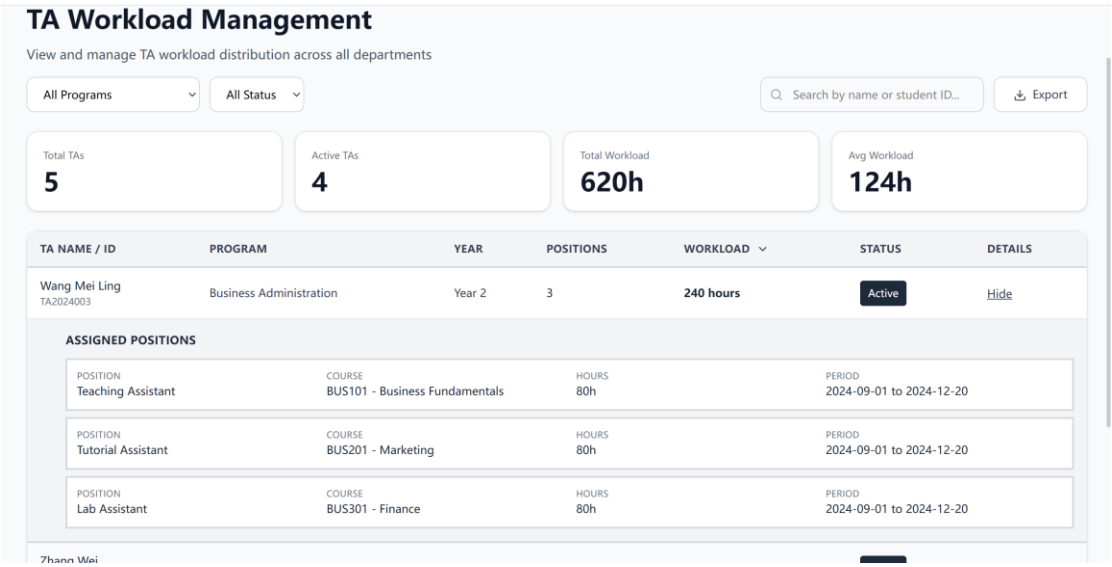
It pre-calculates core statistical results, such as the total number of TAs, total job openings, recruitment completion rate, and the approximate workload of each TA.

It presents these data through concise cards and charts on the dashboard, allowing the administrator to grasp the overall system status at a glance and navigate directly to more detailed views.

(2) Checking TA Workload



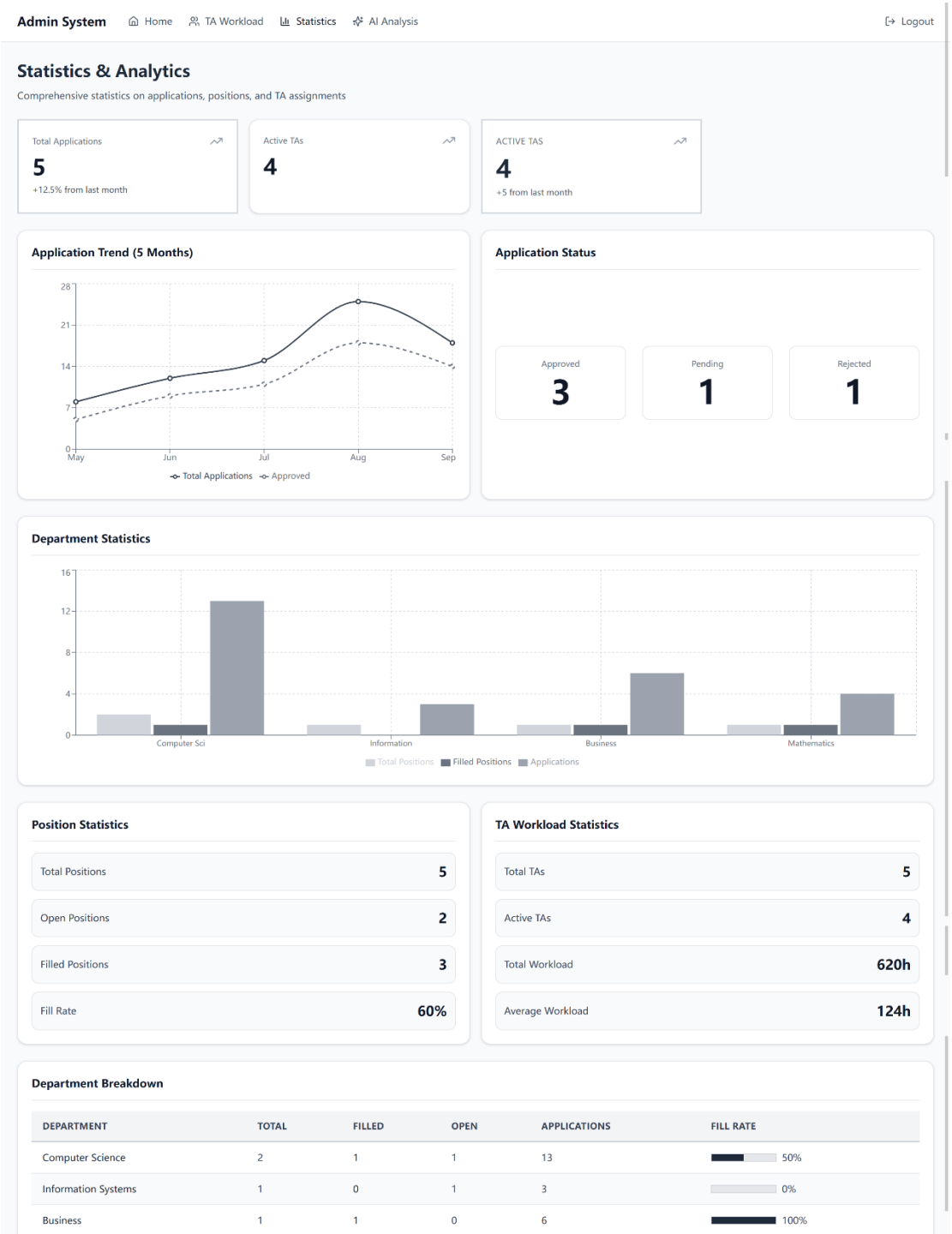
When the administrator needs to understand the exact workload assigned to each TA, they can click to enter the **TA Workload Page** from the dashboard.



The system reorganizes the data based on TA names, affiliated departments, and time periods. It compiles a clear breakdown of each TA's assigned job count, processed applications, and task completion status, then displays this information in structured lists and visual charts. The administrator can use filtering tools to quickly locate specific TAs, clearly identifying who is overloaded or underutilized, and use

this insight to adjust job allocations accordingly.

(3) Reviewing Job Allocation and Overall Data



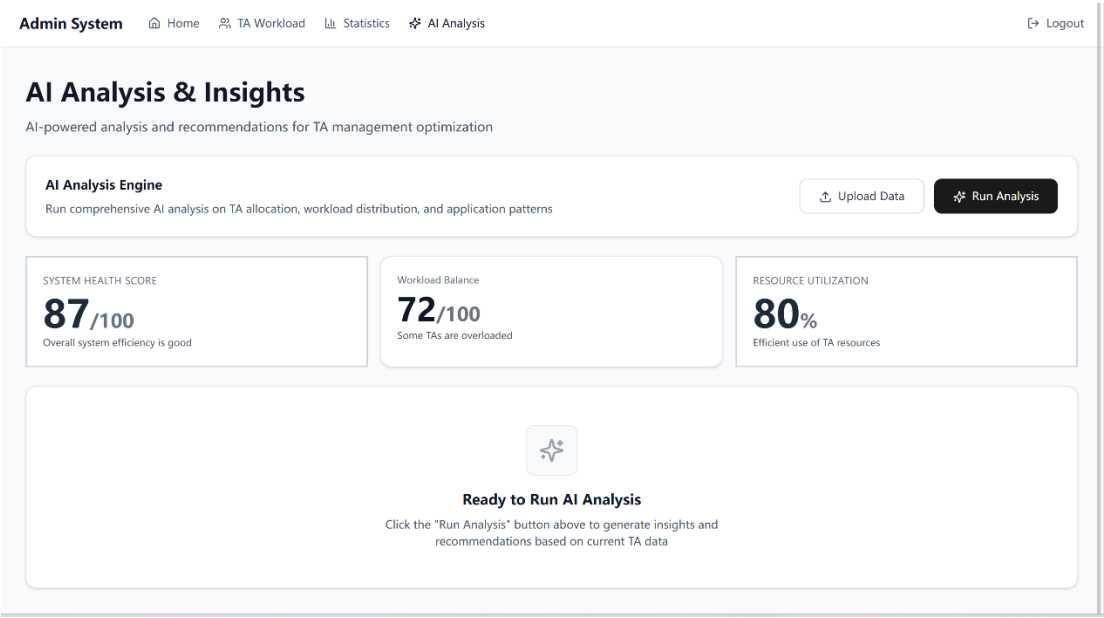
If the administrator wants to focus on job allocation and the overall recruitment progress, they can navigate to the **Statistics Page** from the dashboard.

The system presents data in two distinct dimensions:

Job Allocation Statistics: It calculates the required headcount for each position, the number of TAs already assigned, recruitment gaps, and the distribution of jobs across departments, clearly illustrating whether job allocation is balanced.

Overall Recruitment Data: It tracks the full recruitment lifecycle from job posting to resume receipt, interviews, offer issuance, and onboarding. It also calculates conversion rates at each stage, helping the administrator identify bottlenecks, inefficiencies, and areas that require additional resources.

(4) Reviewing Job AI Expansion Process



This module has pre-built entry points for AI analysis, which will simplify operations once the corresponding features are launched:

The administrator clicks the corresponding function button on the **AI Analysis Entry Page**. The system automatically attaches the current filtering criteria (such as time range, department, or TA name) to call the AI interface. Once the AI analysis is completed, the results are displayed directly on the page, including TA skill matching scores, suitable job recommendations, and workload optimization suggestions. The administrator can directly apply these insights to adjust job allocations or TA work arrangements, making management more efficient.